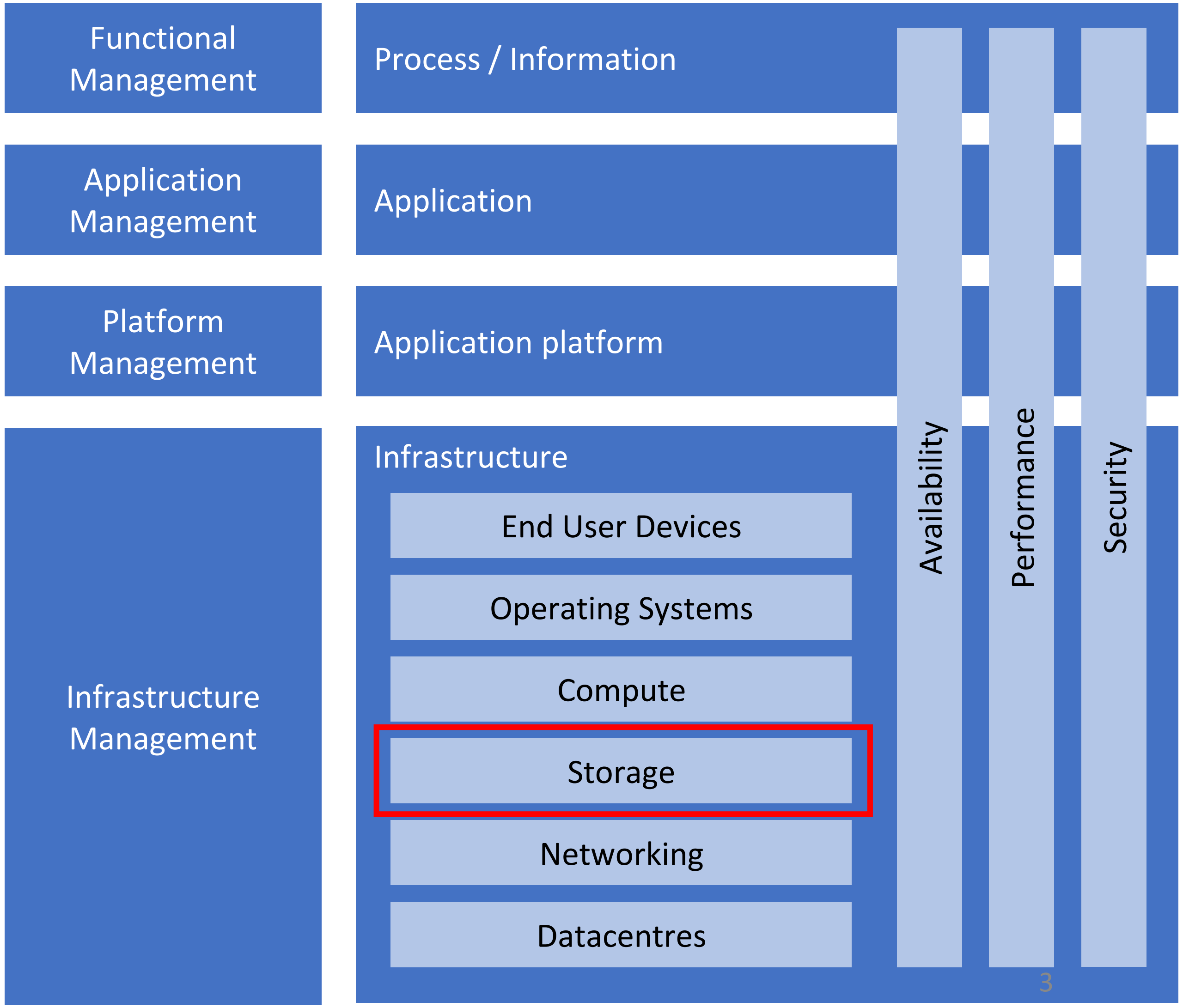
Chapter 5: Storage

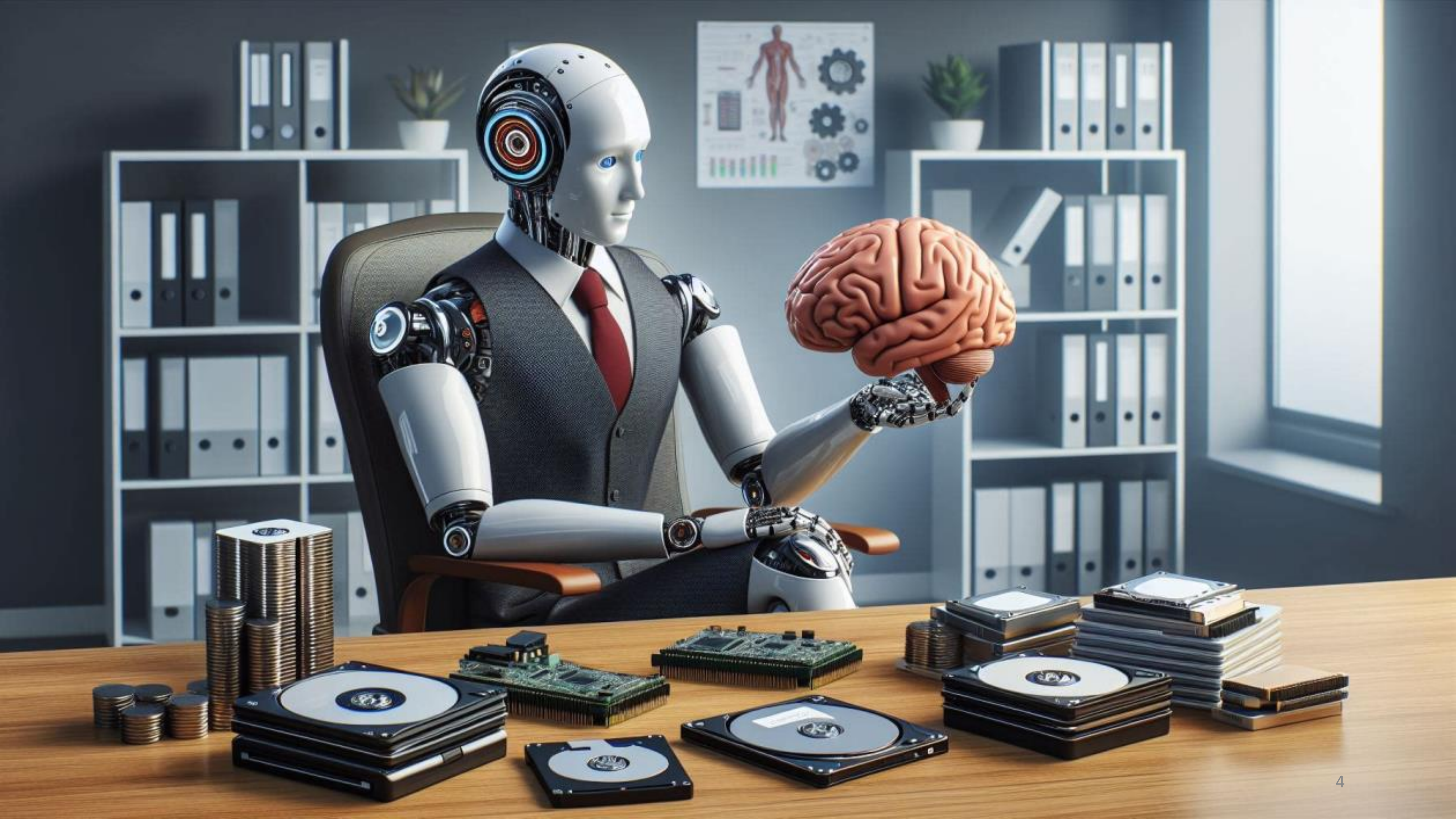
Objectives

- Discuss the storage building blocks
- Discuss storage availability
- Discuss storage performance
- Discuss storage security

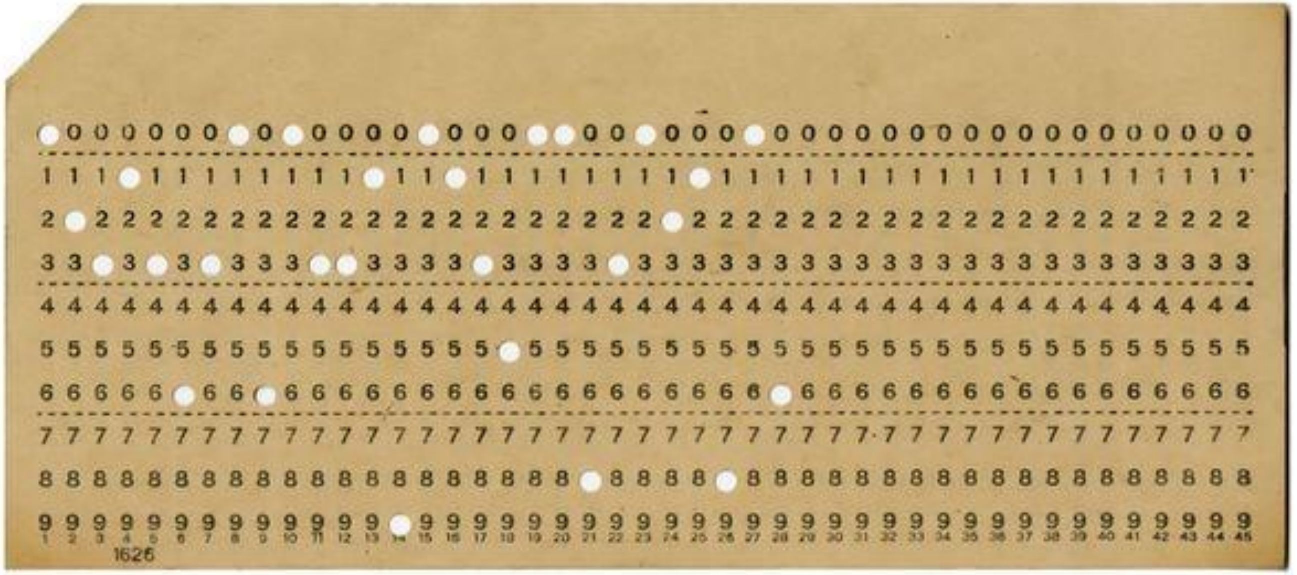
Introduction

- Every day, approximately 200,000 petabytes of new information is generated worldwide
- The total amount of digital data doubles approximately every two years

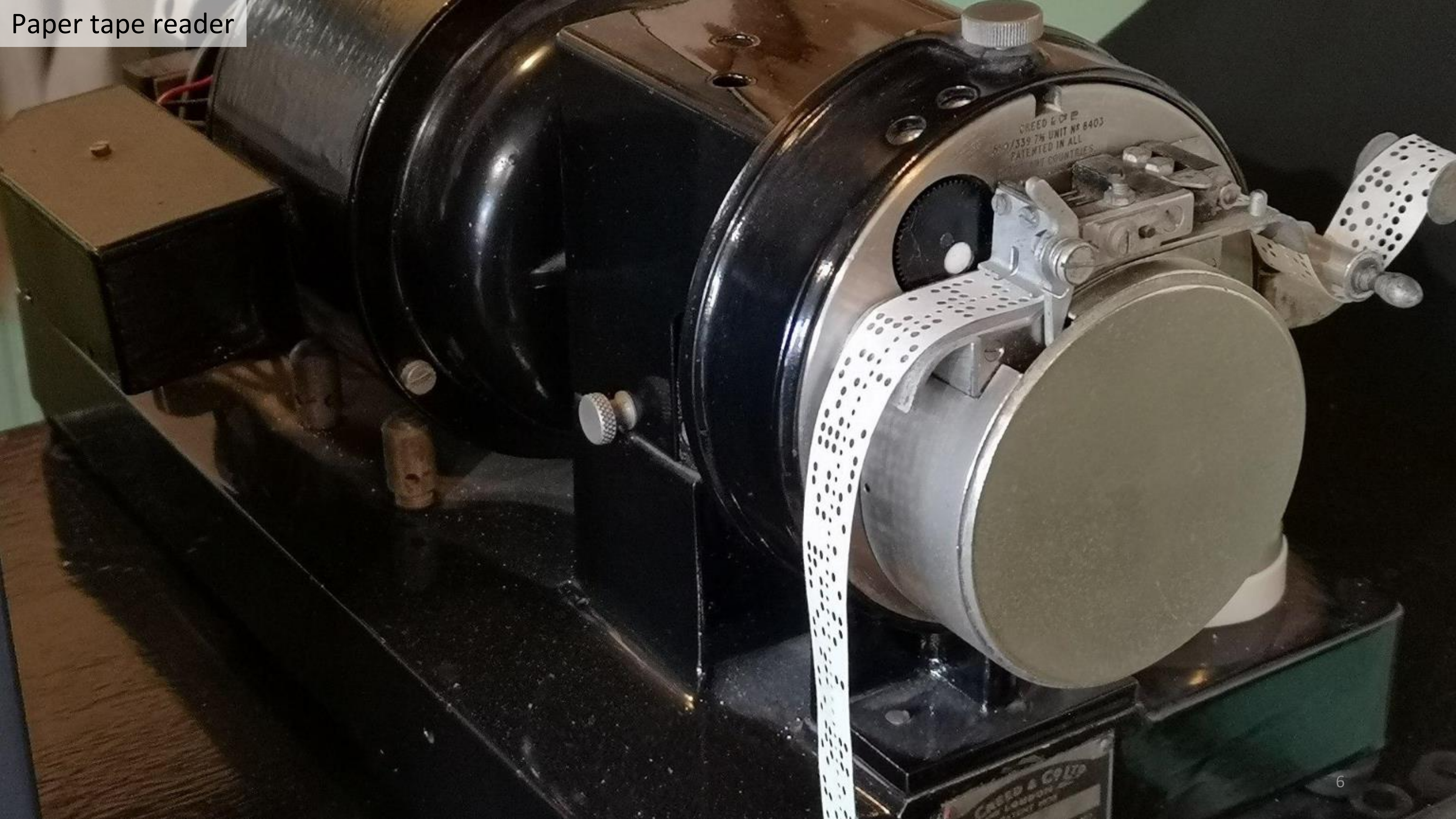




Punch Card

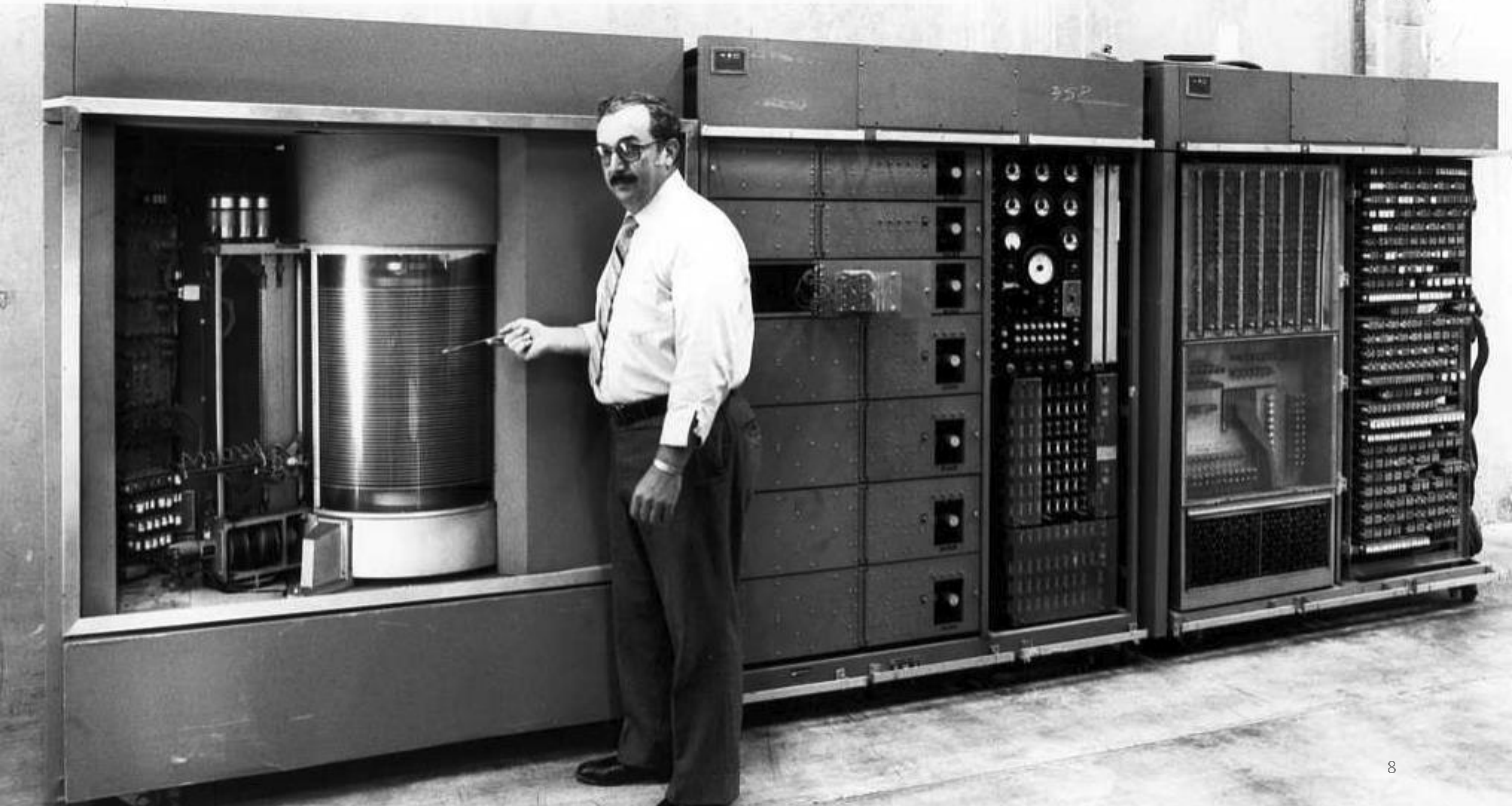


Paper tape reader



Drum memory (Size 62.5 kilobytes)

the IBM RAMAC 350 – first computer with a hard disk (Size 5 megabytes)



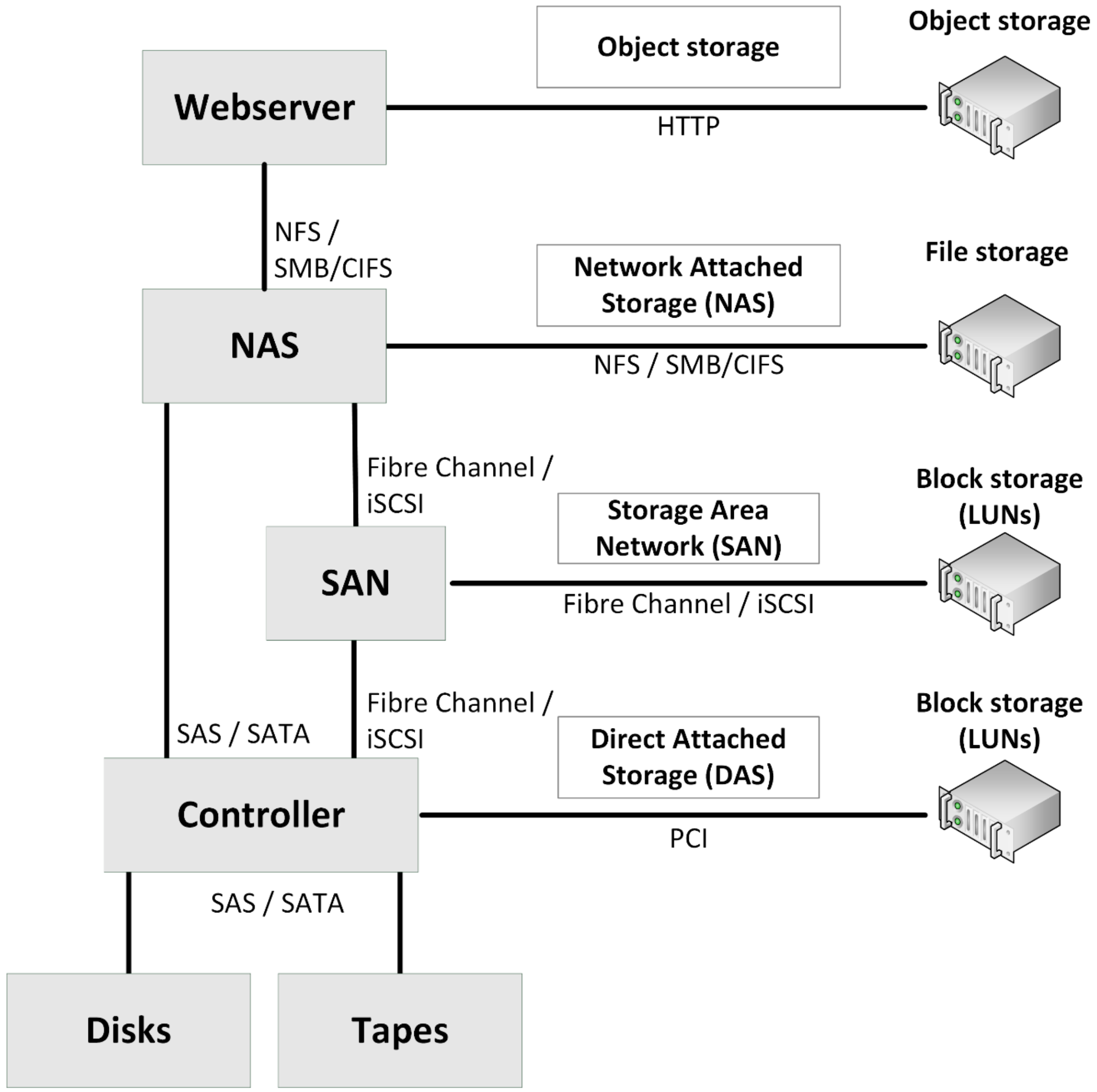
IBM 726 (Size 1 MB per 10 cm)

Digital Linear Tape (Size: 300 GB)

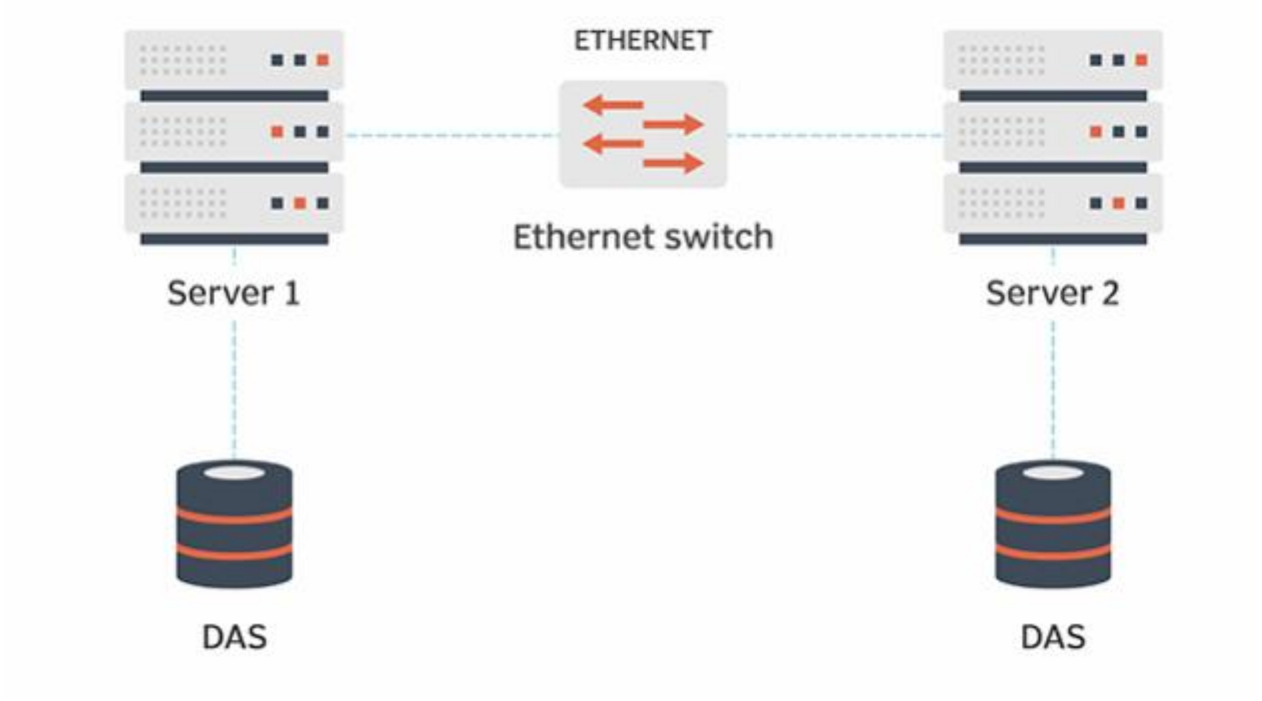
Linear Tape Open (Size: 6 TB)

Storage building blocks

Storage model



Direct Attached Storage (DAS)

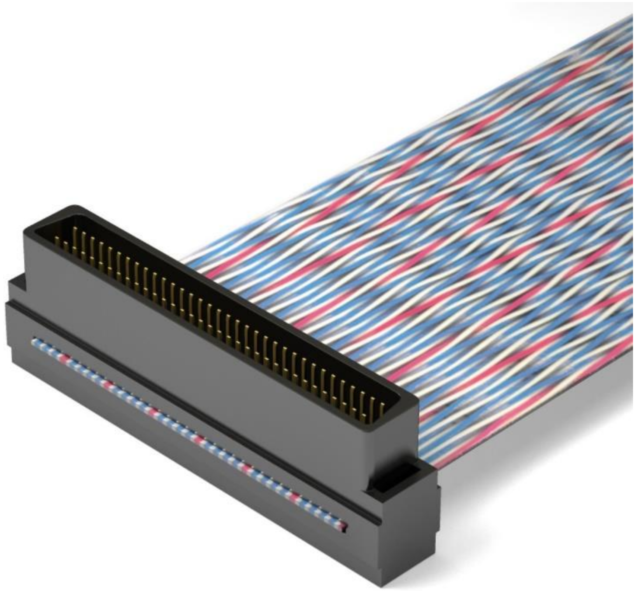


Disks – Command Set

Disks – command sets

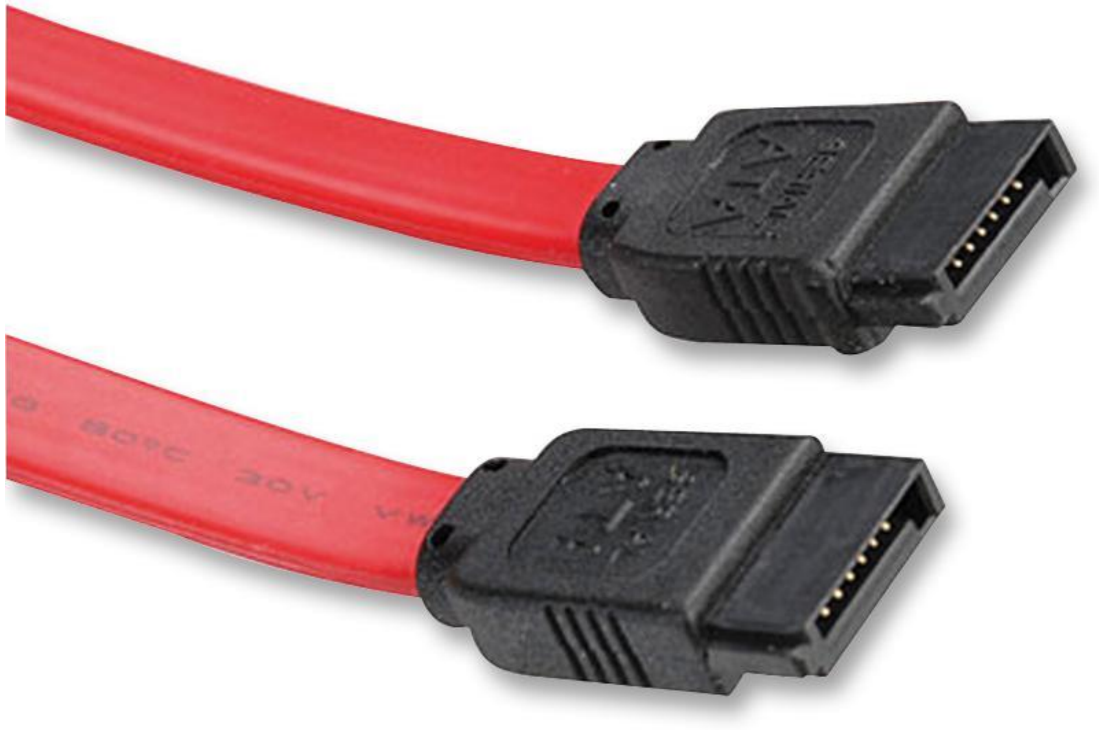
- Disks are connected to disk controllers using a command set.

Advanced Technology Attachment (ATA) or
Integrated drive electronics (IDE)



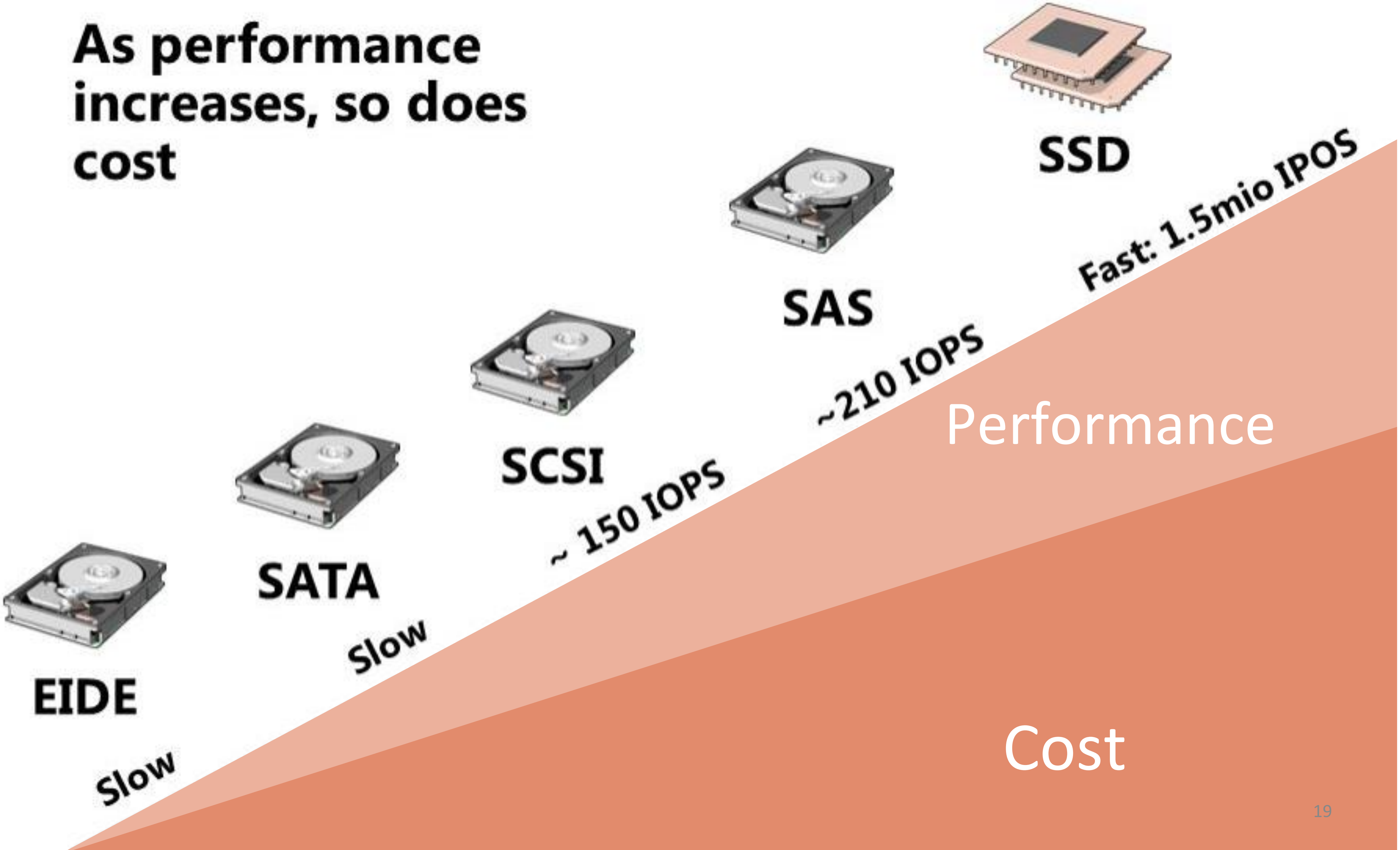
Small Computer System Interface (SCSI)

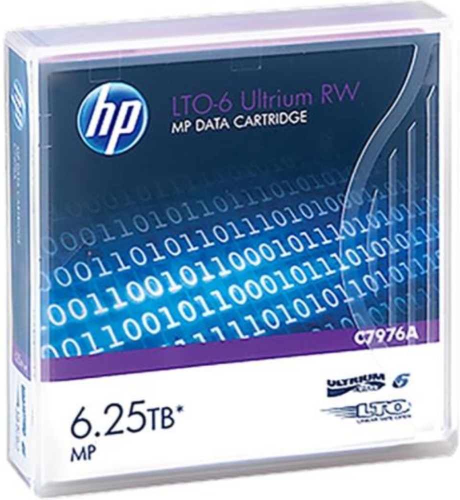
Disks – command sets



Serial Advanced Technology Attachment (SATA)

As performance increases, so does cost





Size: 6.25 TB
MYR 350

Size: 3 TB
MYR 780

Size: 960 MB
MYR 320

Size: 1 TB
MYR 500

Mechanical hard disks

Mechanical hard disks

- Serial ATA (SATA) disks

Mechanical hard disks

- Near-Line SAS (NL-SAS) disks
  - Have a SAS interface, but the mechanics of SATA disks
  - Can be combined with faster SAS disks in one storage array

Mechanical hard disks

Serial Attached SCSI (SAS) disks

Solid State Drives (SSDs)

SATA
SSD

M.2 SSD

Solid State Drives (SSDs)

Advantages:

1. Consume less power

1. Generate less heat

1. No moving parts - no vibrations

1. Prices reduced

Solid State Drives (SSDs)

Disadvantages:

- Life expectancy

- Slow compared to Nonvolatile memory express (NVMe) drives

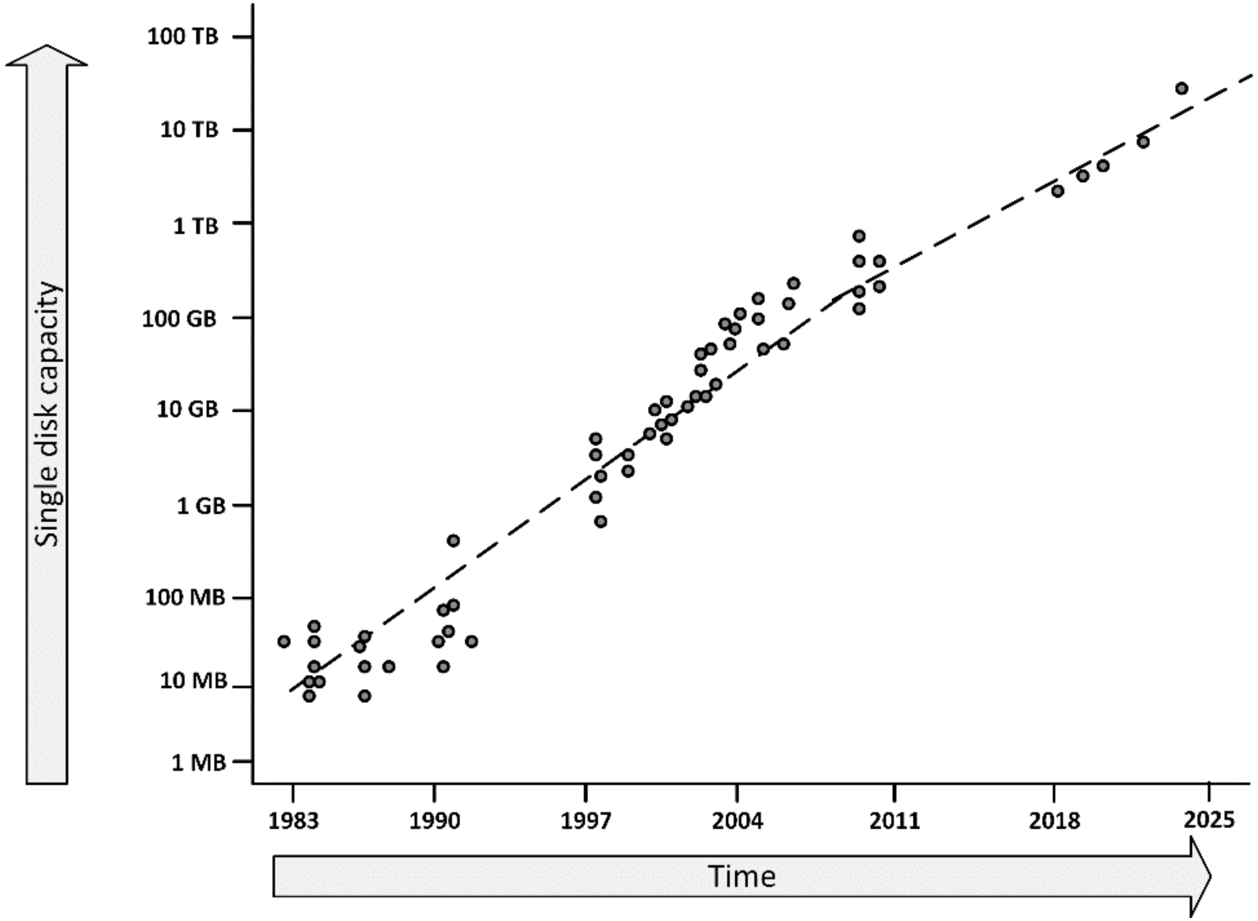
NVMe SSD

M.2 SSD

SSD vs HDD

SSD			HDD
faster	✓	✗	slower
more expensive	✗	✓	cheaper
non-mechanical (flash)	✓	✗	mechanical (moving parts)
shock-resistant	✓	✗	fragile
best for storing operating systems, gaming apps, and frequently used files			best for storing extra data, such as movies, photos, and documents

Disk capacity - Kryder's law



Please note that the vertical scale is logarithmic instead of linear

Disk capacity - Kryder's law

- To have full benefits of Kryder's law, the storage infrastructure should be designed to **handle just in time expansion**

2004 2014 2024

YEAR

Tapes

- Large amounts of data
- Inexpensive
- Tapes are suitable for archiving - long life expectancy

Tapes - disadvantages

- Tapes are fragile
- Tape cartridges contain mechanical parts
- Frequent rewinding causes stress to the tape substrate. Leads to lower reliability of data reads

NAME	FACTOR	VALUE IN BYTES
kilobyte	10^3	1,000
megabyte	10^6	1,000,000
gigabyte	10^9	1,000,000,000
terabyte	10^{12}	1,000,000,000,000
petabyte	10^{15}	1,000,000,000,000,000
exabyte	10^{18}	1,000,000,000,000,000,000
zettabyte	10^{21}	1,000,000,000,000,000,000,000
yottabyte	10^{24}	1,000,000,000,000,000,000,000,000

Tapes

- (S)DLT and LTO are the most popular tape cartridge formats in use today
  - LTO has a market share of more than 80%
  - LTO-9 tape cartridges can store 18 TB of uncompressed data

- Tape throughput is in the 200 MB/s range
  - The tape drive interface is capable of even higher speeds
  - Most tape drives use 8 Gbit/s Fibre Channel interfaces. A sustained throughput of between 700 and 900 MB/s

Tape library

- Tape libraries can be used to automate tape handling

- A tape library is a storage device that contains:
  - One or more tape drives
  - A number of slots to hold tape cartridges
  - A barcode or RFID tag reader to identify tape cartridges
  - An automated method for loading tapes


BM TS4500 Tape Library - Dual Robotics In Action

Virtual tape library

- A Virtual Tape Library (VTL) uses disks for storing backups
- A VTL consists of:
  - An appliance or server
  - Software that emulates traditional tape devices and formats
- VTLs combine high performance disk based backup and restore with well-known backup applications, standards, processes, and policies
- Most of the current VTL solutions use NL-SAS or SATA disk arrays because of their relatively low cost
- They provide multiple virtual tape drives for handling multiple tapes in parallel

Feature	HDD	SSD	Tape
Performance	Slower (moving parts, rotational latency)	Faster (instantaneous access, no moving parts)	Slowest (sequential access, needs rewind)
Capacity	Highest capacity options available	Lower capacity than HDDs at comparable price	Highest theoretical capacity
Cost per GB	Lower cost per GB	Higher cost per GB	Very low cost per GB
Reliability	More susceptible to physical damage	Less susceptible to physical damage	Highly durable, long archival lifespan

Controllers

- Controllers connect disks and/or tapes to a server, in one of two ways:
  - Implemented as a PCI expansion boards in the server
  - Part of a Network Attached Storage (NAS) or Storage Area Network (SAN) deployment, where they connect all available disks and tapes to redundant Fibre C onnec

PCI expansion Disk Controller

Network attached storage (NAS)

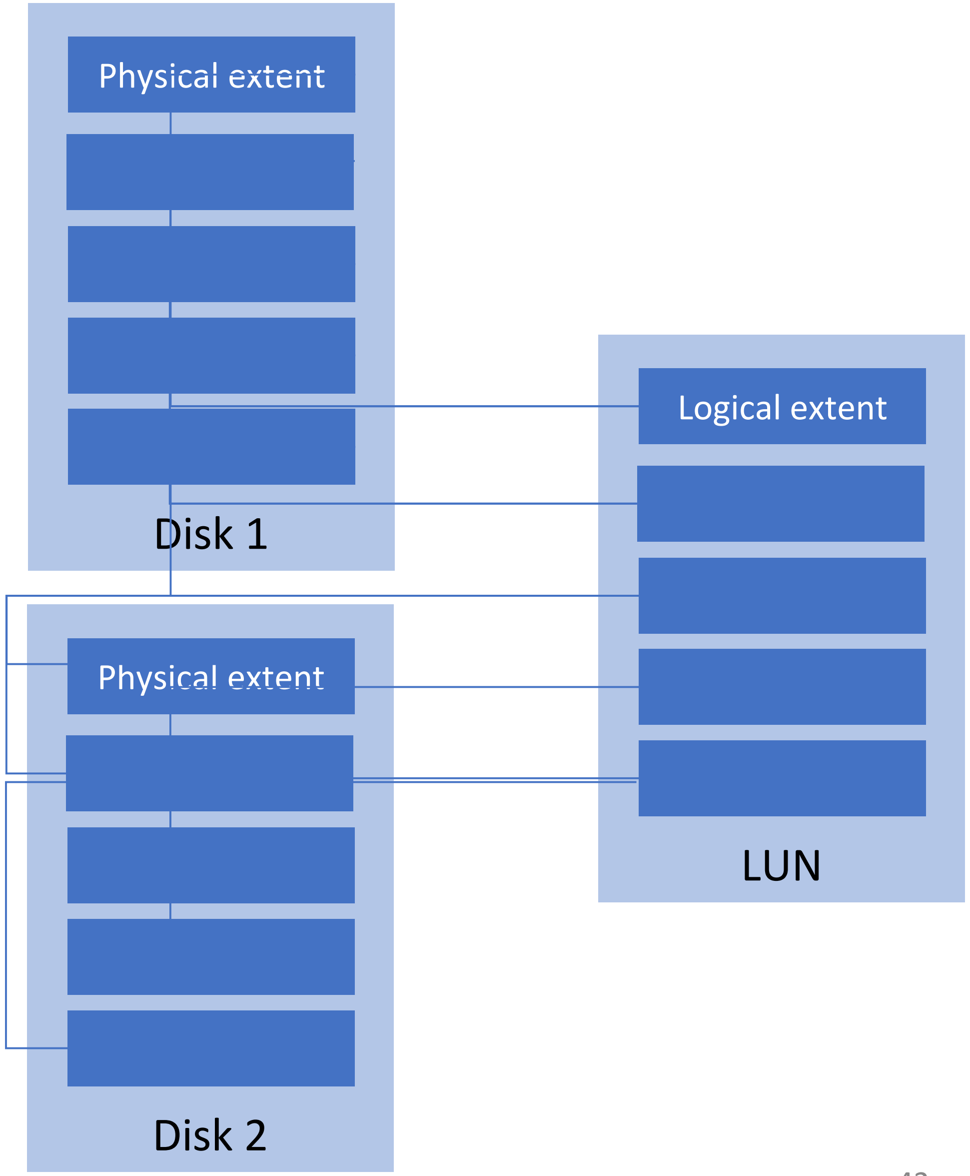
Controllers

- A controller can implement:
  - High performance
  - High availability
  - Virtualized storage
  - Cloning
  - Data deduplication*
  - Thin provisioning*

* Will be discussed

Controllers

- The controller splits up all disks in small pieces called physical extents
- From these physical extents, new virtual disks (Logical Unit Numbers - LUNs) are composed and presented to the operating system



RAID

- Redundant Array of Independent Disks (RAID) solutions provide:
  - High availability of data
  - Improvements of performance
- RAID uses multiple redundant disks
- RAID can be implemented:
  - In the disk controller’s hardware
  - As software running in a server’s operating system

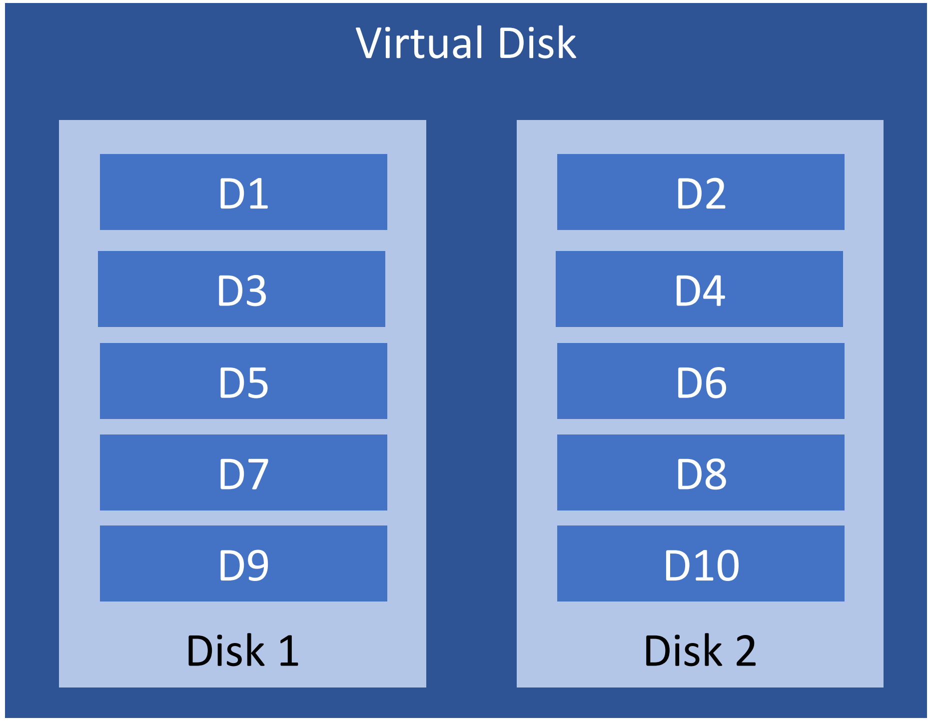
RAID

- RAID can be implemented in several configurations, called RAID levels

- In practice, five RAID levels are implemented most often:
  - RAID 0 - Striping
  - RAID 1 - Mirroring
  - RAID 10 - Striping and Mirroring
  - RAID 5 - Striping with distributed parity
  - RAID 6 - Striping with distributed double parity

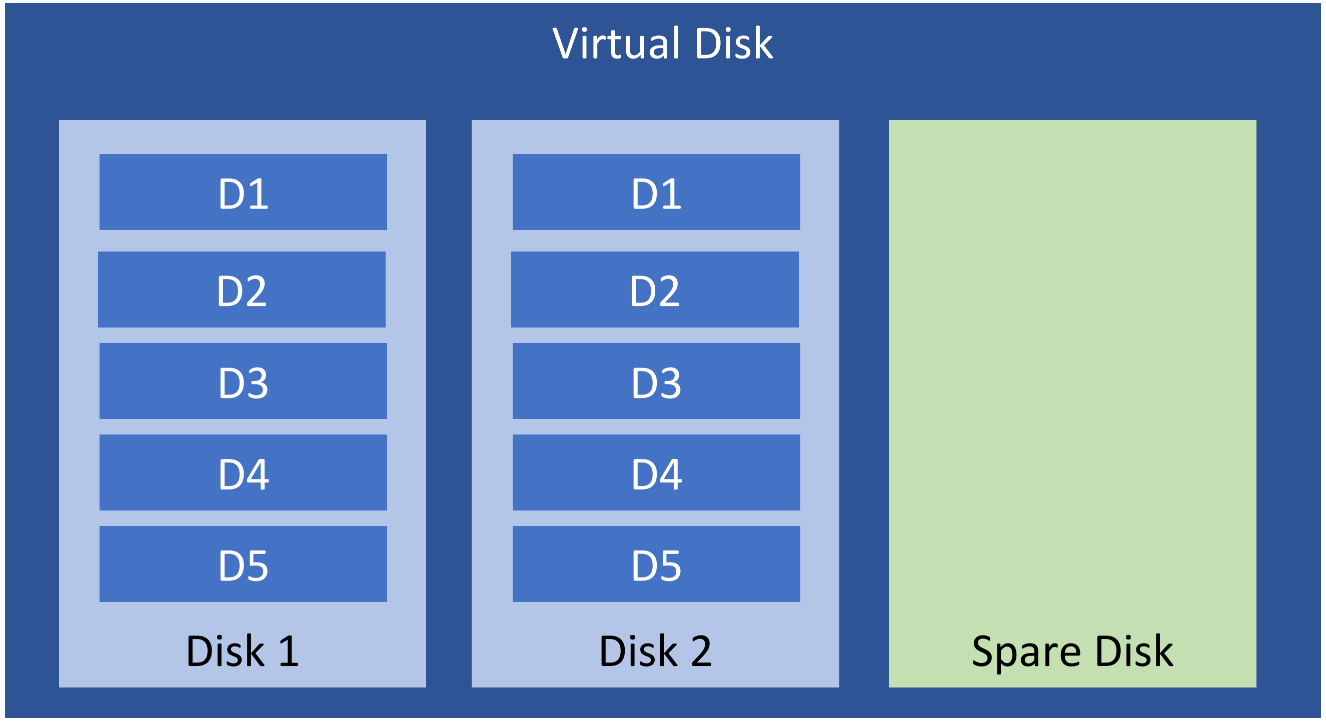
RAID 0 - Striping

- **Easy** and **cheap**
- **High performance**: uses multiple disks, each with a part of the data on it
- **Lowers availability :** losing all data is acceptable, for instance for temporary data



RAID 1 - Mirroring

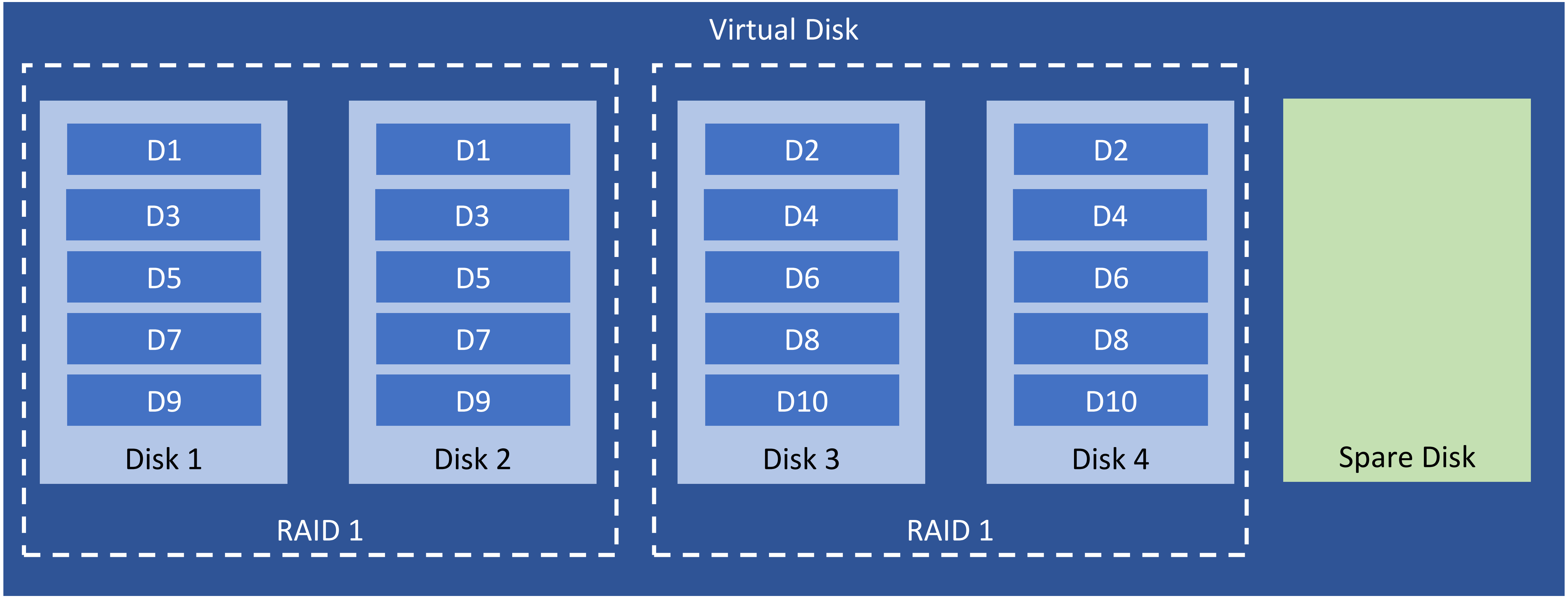
- High availability - redundancy
- The most reliable RAID level
- Expensive: high price
- Has a spare physical disk



RAID 10 - Striping and mirroring

- RAID 10 uses a combination of striping and mirroring
- Provides **high performance** and **availability**
- Only 50% of the available disk space is used

RAID 10 - Striping and mirroring



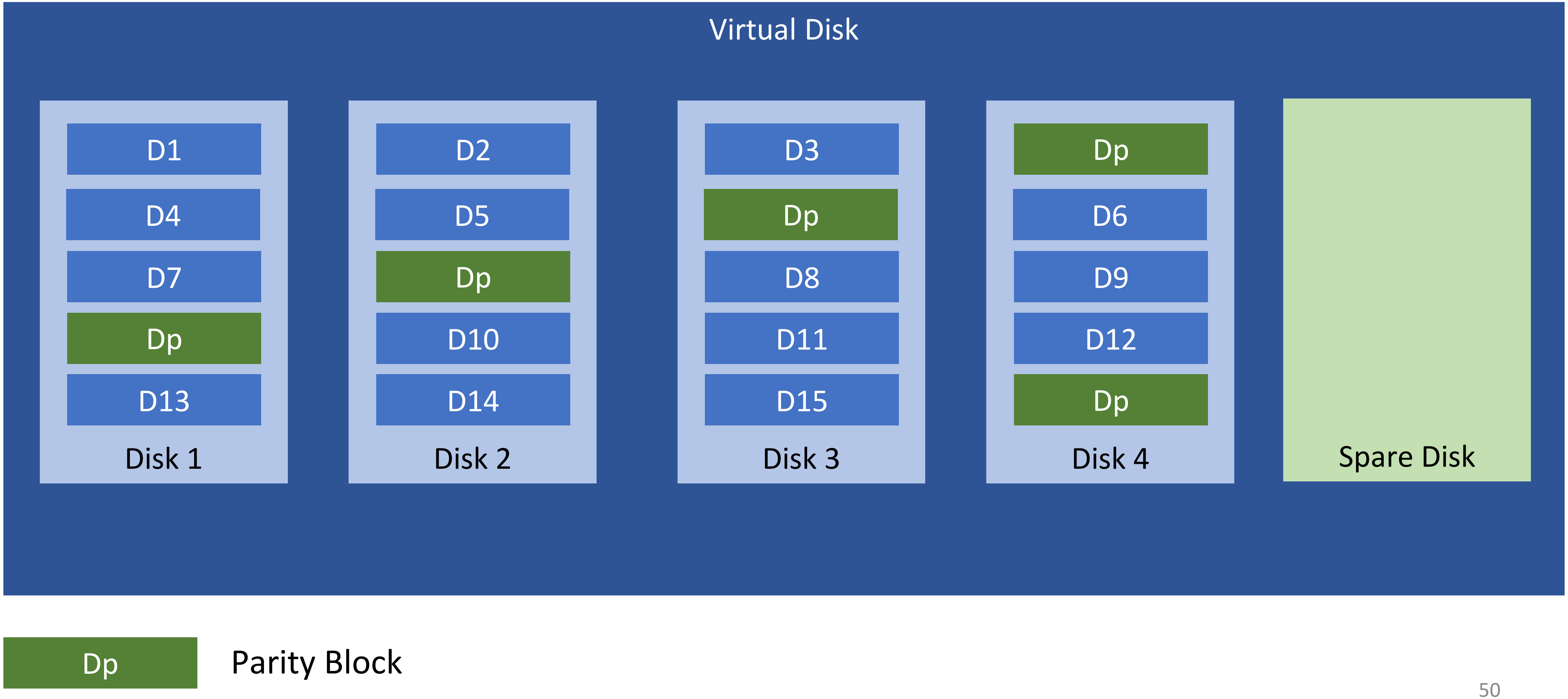
RAID 5 - Striping with distributed parity

- Data is written in disk blocks on all disks

- A **parity block** of the written disk blocks is stored as well

- This **parity block** is used to automatically **reconstruct data** in a RAID 5 set (using a spare disk) in case of **a disk failure**

RAID 5 - Striping with distributed parity

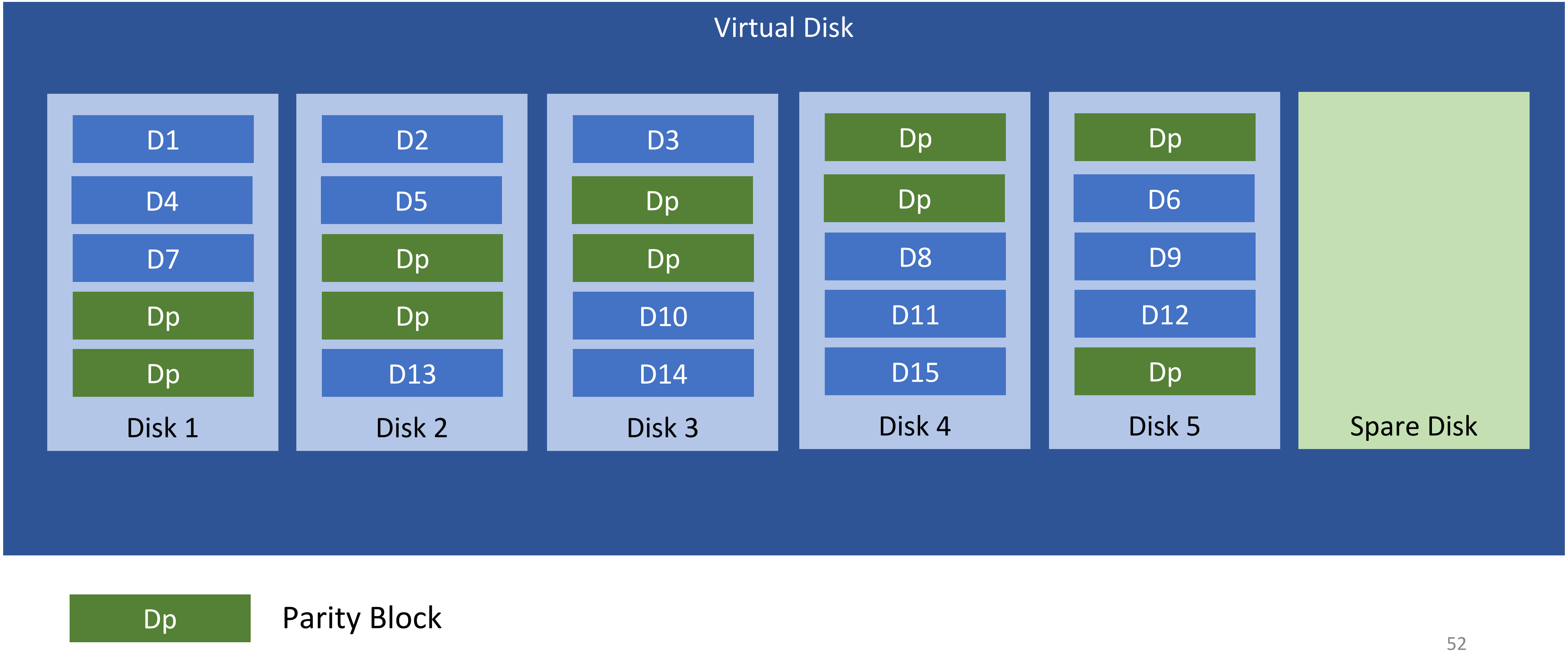


RAID 6 - Striping with distributed double parity

- RAID 6 **protects** against **double disk failures** by using **two distributed parity blocks** instead of one

- Important in case a second disk fails during reconstruction of the first failing disk

RAID 6 - Striping with distributed double parity



Data compression

- Data on disk and tape is typically stored in a **compressed format**

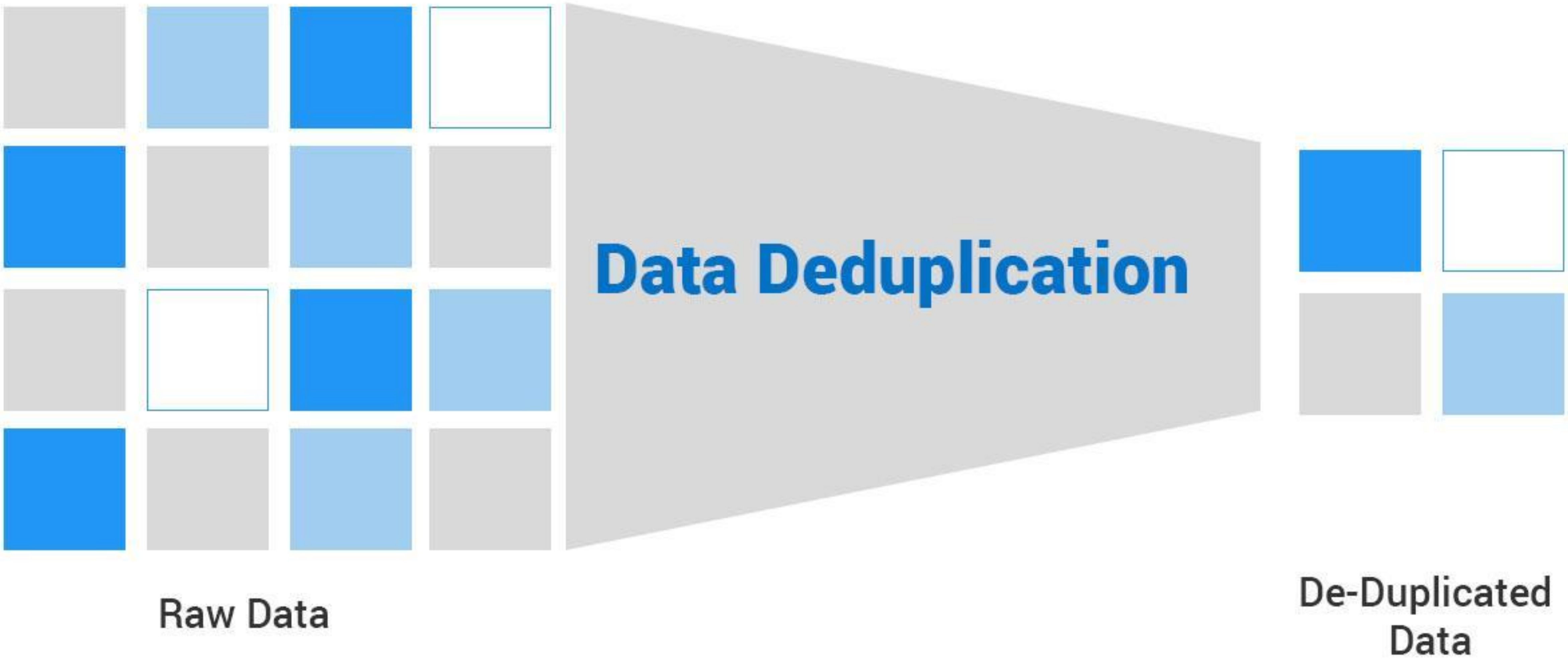
- Allows 2 to 2.5 times the amount of data to be stored on the same media

- The degree of compression is never guaranteed - if the data is very diverse, the compression ratio may be correspondingly lower

Data deduplication

- Data deduplication searches the storage system for **duplicate data** segments (disk blocks or files) and **removes** these duplicates

- Data deduplication is used in archived as well as in production data

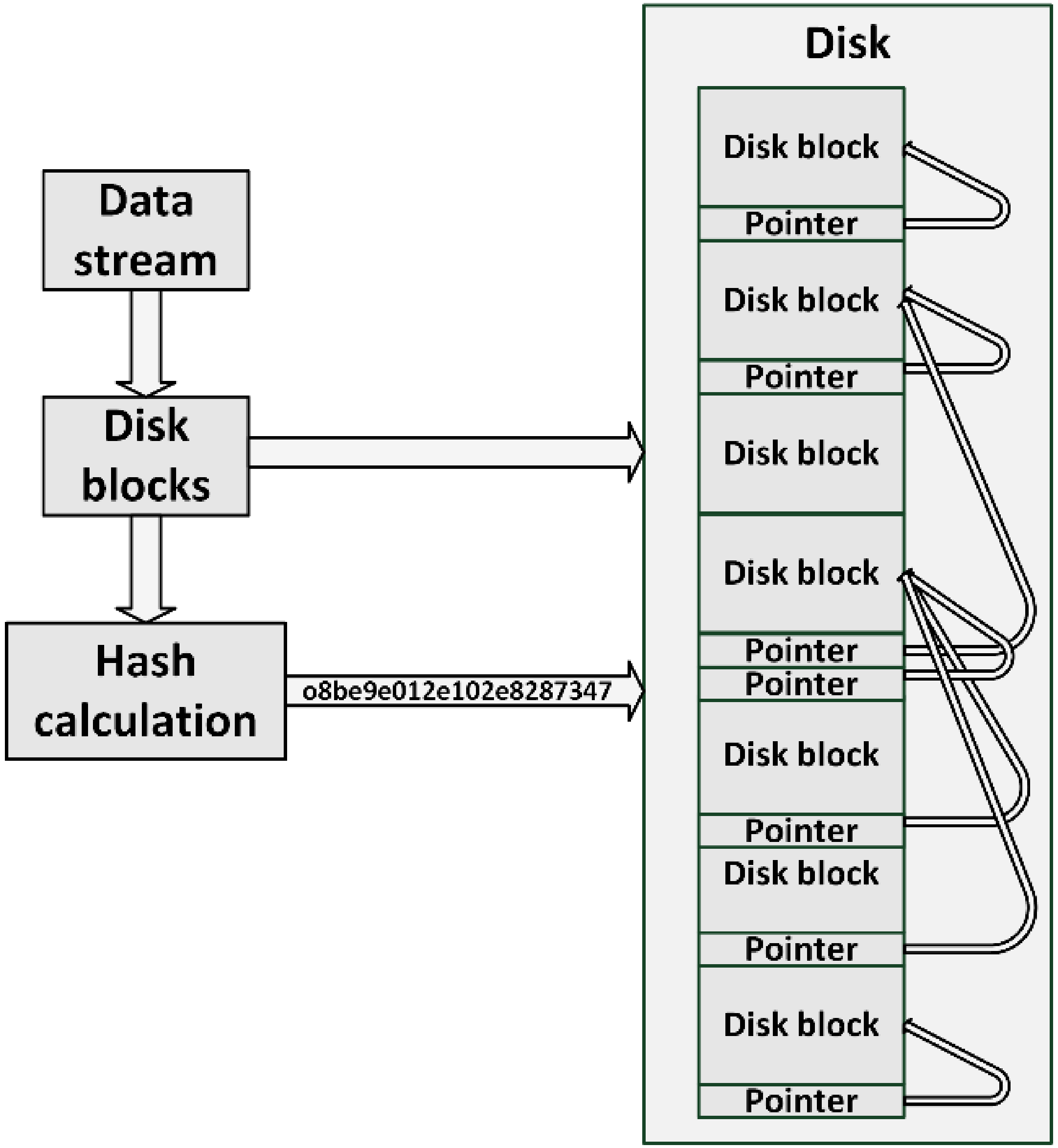

Data Deduplication
Raw Data
De-Duplicated
Data

Data deduplication

- The deduplication system keeps a table of hash tags to quickly identify duplicate disk blocks
  - The incoming data stream is segmented
  - Hash tags are calculated of those segments
  - The hashes are compared to hash tags of segments already on disk
  - If an incoming data segment is identified as a duplicate, the segment is not stored again, but a pointer to the matching segment is created for it instead

Data deduplication

- Deduplication can be done inline or periodically
- Inline deduplication checks for duplicate data segments before data is written to disk
  - Avoids duplicate data on disks at any time
  - Introduces a relatively large performance penalty



Data deduplication

- Periodically: writing data to disk first, and periodically check if duplicate data exists
  - Duplicate data is deduplicated by changing the duplicate data to a pointer to existing data on disk, and freeing disk space of the original block
  - This process can be done at times when performance needs are low
  - Duplicate data will be stored on the disks for some time

Cloning and snapshots

- With cloning and snapshotting, a copy of data is made at a specific point in time that can be used independently from the source data

- Usage:
  - Create a backup at a specific point in time, when the data is in a stable, consistent state
  - Creating test sets of data and an easy way to revert to older data without restoring data from a backup

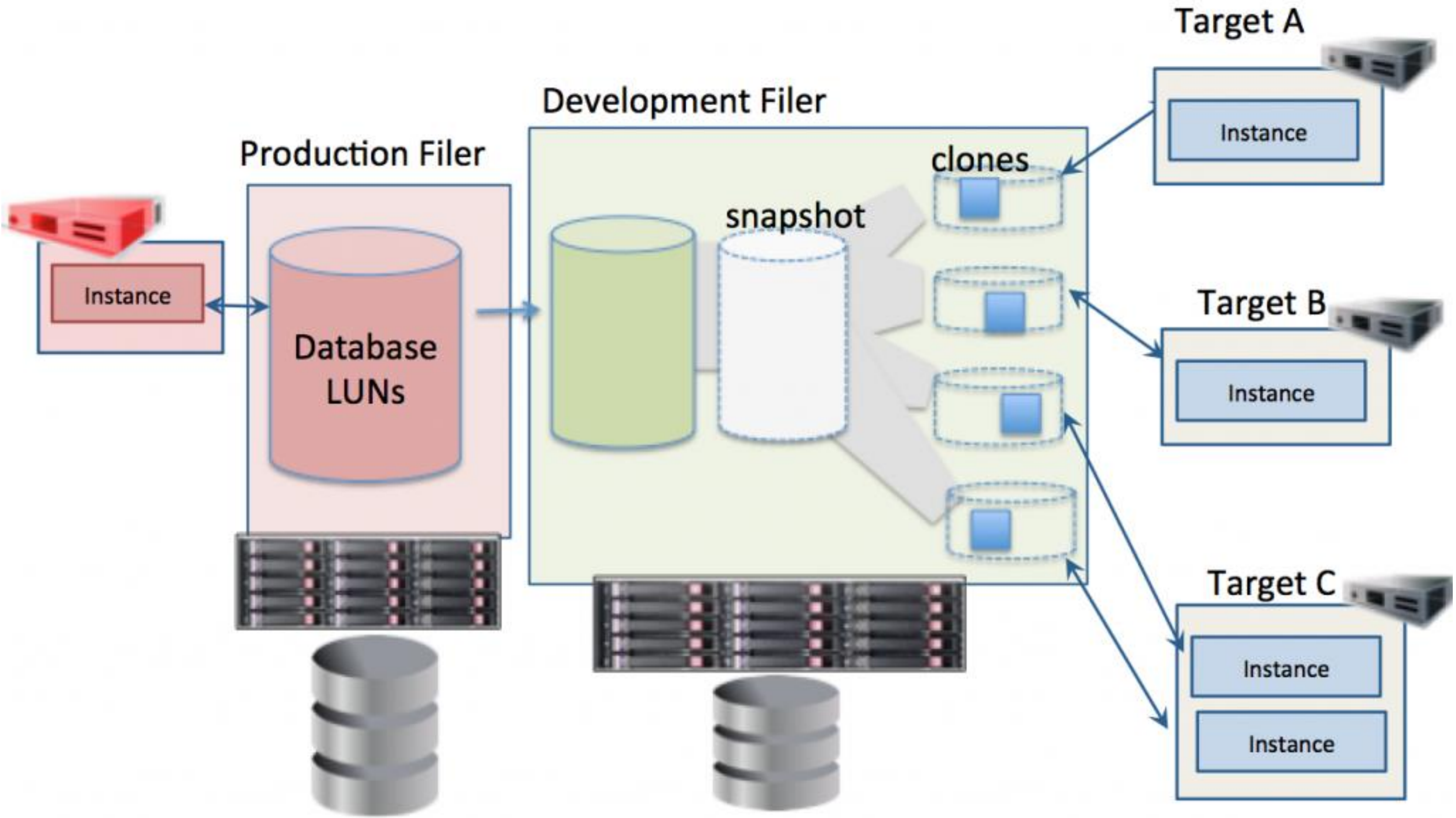
Cloning and snapshots

Cloning

- The storage system creates a full copy of a disk, much like a RAID 1 mirror disk

Snapshot

- Represents a point in time of the data on the disks
  - No writing to those disks is permitted anymore, as long as the snapshot is active
  - All writing is done on a separate disk volume in the storage system
  - The original disks still provide read-access


Target A
Development Filer
Instance
Production Filer
clones
snapshot
Instance
Database
LUNs
Target B
Instance
Target C
Instance
Instance

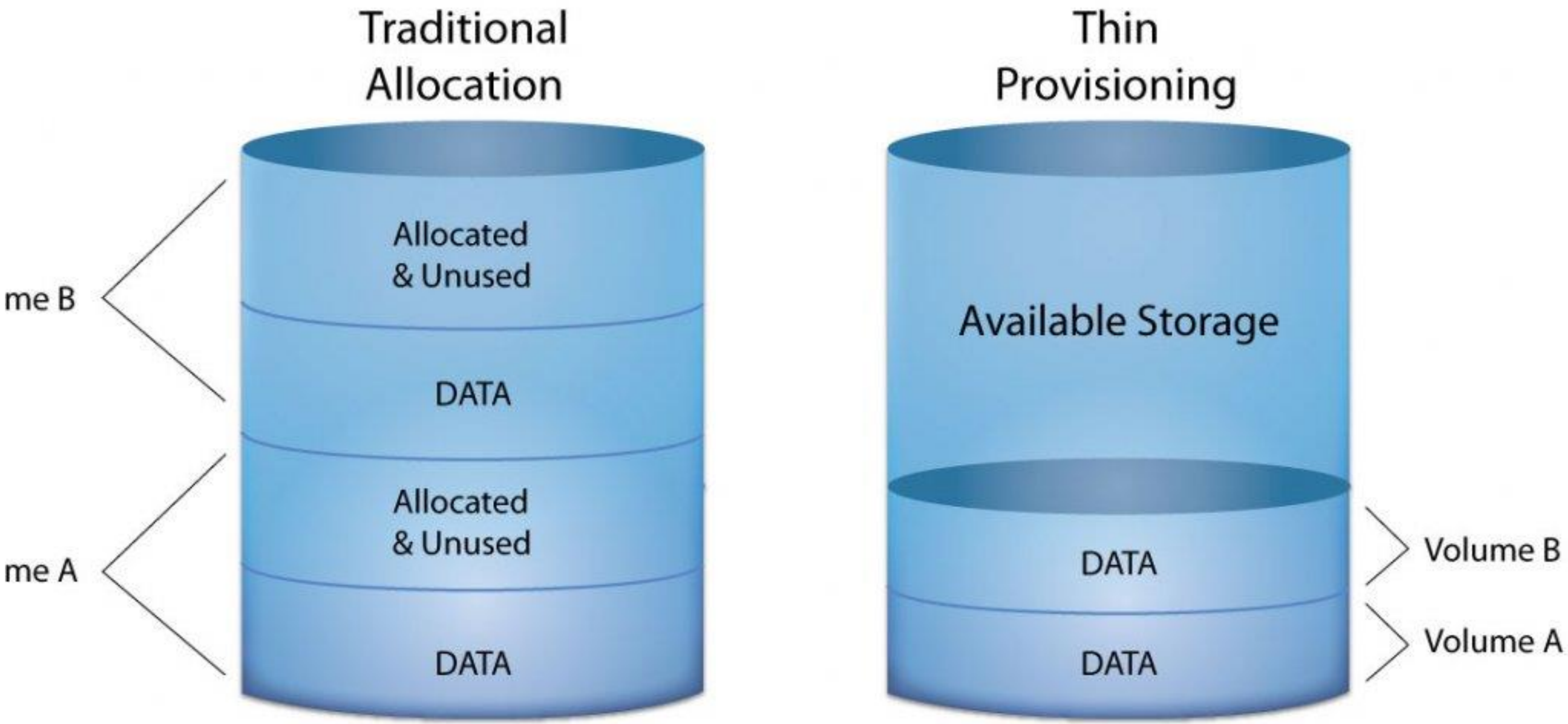
Feature	**Cloning**	**Snapshots**
Scope	Full copy	Point-in-time image
Time-consuming	Yes	No
Reversibility	Complex	Easy
Use cases	Creating new VMs, restoring from backups	Backups, testing, recovery

Thin provisioning

- Thin provisioning enables the **allocation of more storage capacity** to users **than is physically installed**. About 50% of allocated storage is never used

- Thin provisioning still provides the applications with the required storage
  - Storage is not really available on physical disks
  - Uses automated capacity management
  - The application's real storage need is monitored closely
  - Physical disk space is added when needed

- Typical use: Providing users with large sized home directories or email storage


Traditional
Allocation
Allocated
& Unused
DATA
Allocated
& Unused
DATA
me B
me A
Thin
Provisioning
Available Storage
DATA
DATA
Volume B
Volume A

Direct Attached Storage (DAS)

- DAS – also known as **local disks** – is a storage system where one or more dedicated disks connect via the SAS or SATA protocol to a built-in controller, connected to the rest of the computer using the PCI bus

- The controller provides a set of disk blocks to the computer, organized in LUNs (or partitions)

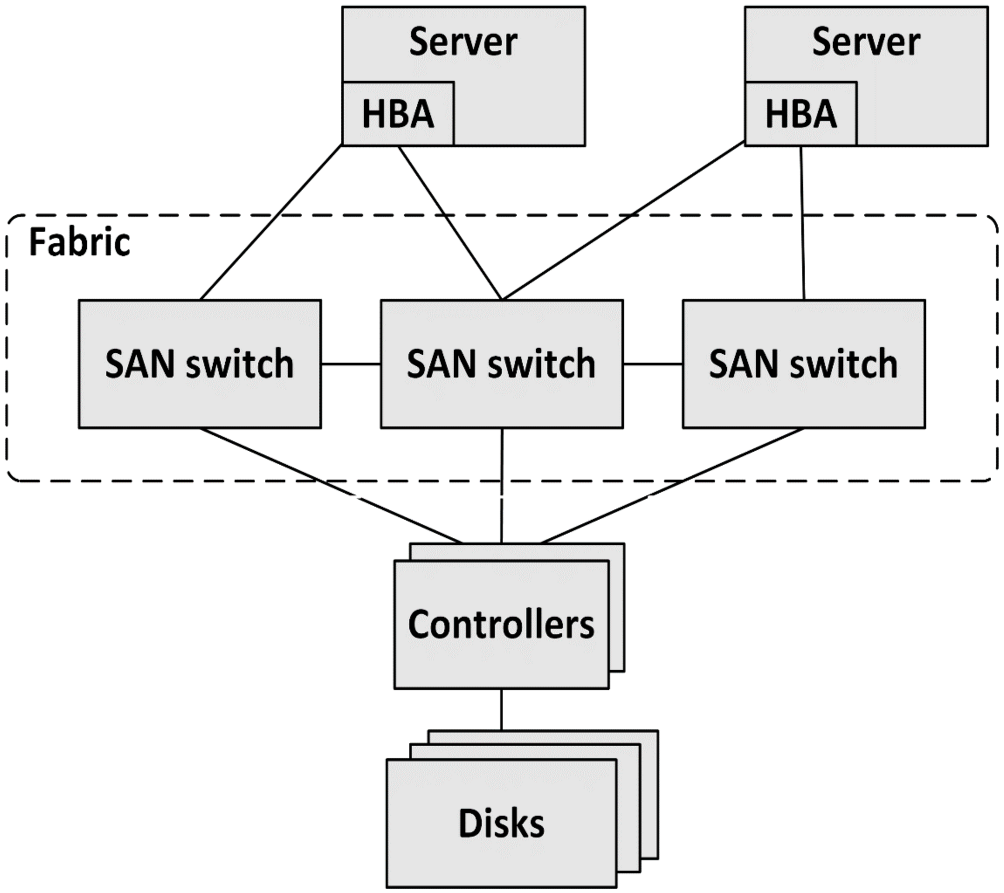
- The computer's operating system uses these disk blocks to create a file system to store files

Storage Area Network (SAN)

- It is a specialized storage network that consists of SAN switches, controllers and storage devices
- It connects a large pool of central storage to multiple servers
- It connects servers to disk controllers using specialized networking technologies like Fibre Channel or iSCSI
- Via the SAN, disk controllers offer virtual disks to servers, also known as LUNs (Logical Unit Numbers)
- LUNs are only available to the server that has that specific LUN mounted

Storage Area Network (SAN)

- The core of the SAN is a set of SAN switches, called the Fabric
- Host bus adapters (HBAs) are interface cards implemented in servers
  - Comparable to NICs used in networking
  - Connected to SAN switches, usually in a redundant way

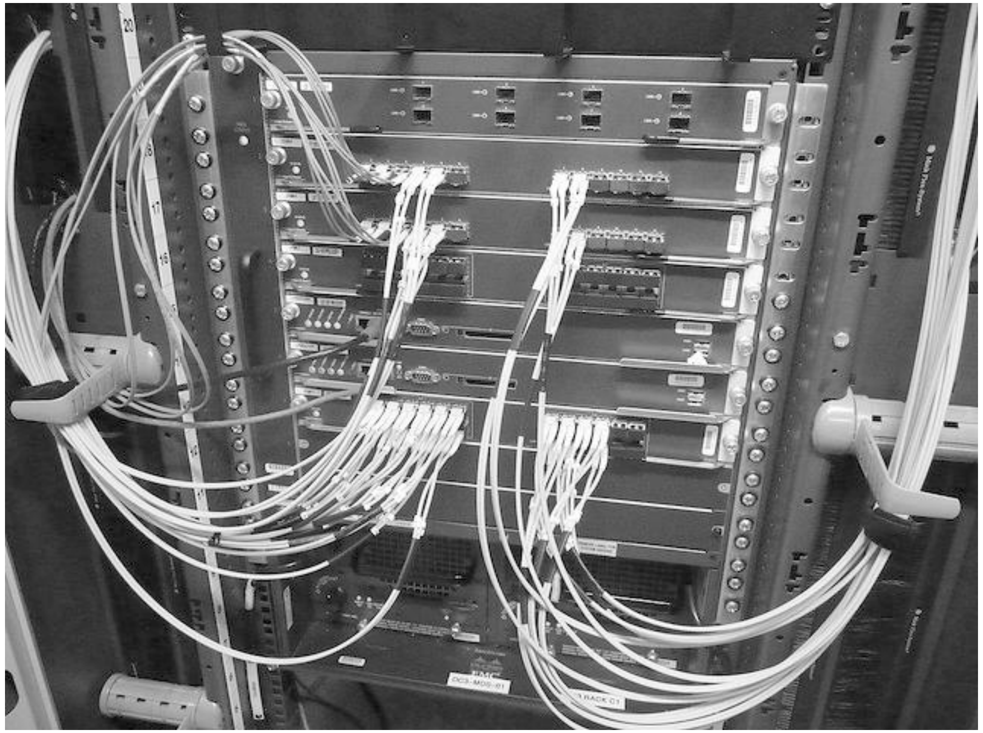


Storage Area Network (SAN)

- A large number of disks are installed in one or more disk arrays
- The number of disks varies between dozens of disks and hundreds of disks
- A disk array can easily contain several petabytes (PB) of data

SAN connectivity protocols

- The most used SAN connectivity protocols:
  - Fibre Channel
  - FCoE
  - iSCSI
- All Fibre Channel devices are connected to Fibre Channel switches, a similar concept as in Ethernet implementations



Fibre Channel

- Fibre Channel (FC) is a dedicated level 2 network protocol, specially designed for transportation of storage data blocks
- Speeds: 1, 2, 4, 8, 16, 32, 64, and 128 Gbit/s
- Runs on:
  - Twisted pair copper wire (i.e. UTP and STP)
  - Fiber optic cables
- The Fibre Channel protocol was specially developed for the transport of disk blocks
- The protocol is very reliable, with guaranteed zero data loss

Fibre Channel

- Three network topologies:
  - Point-to-Point
    - Two devices are connected directly to each other
  - Arbitrated loop
    - Also known as FC-AL
    - All devices are in a loop
  - Switched fabric
    - All devices are connected to Fibre Channel switches
    - A similar concept as in Ethernet implementations
- Most implementations today use a switched fabric

FCoE

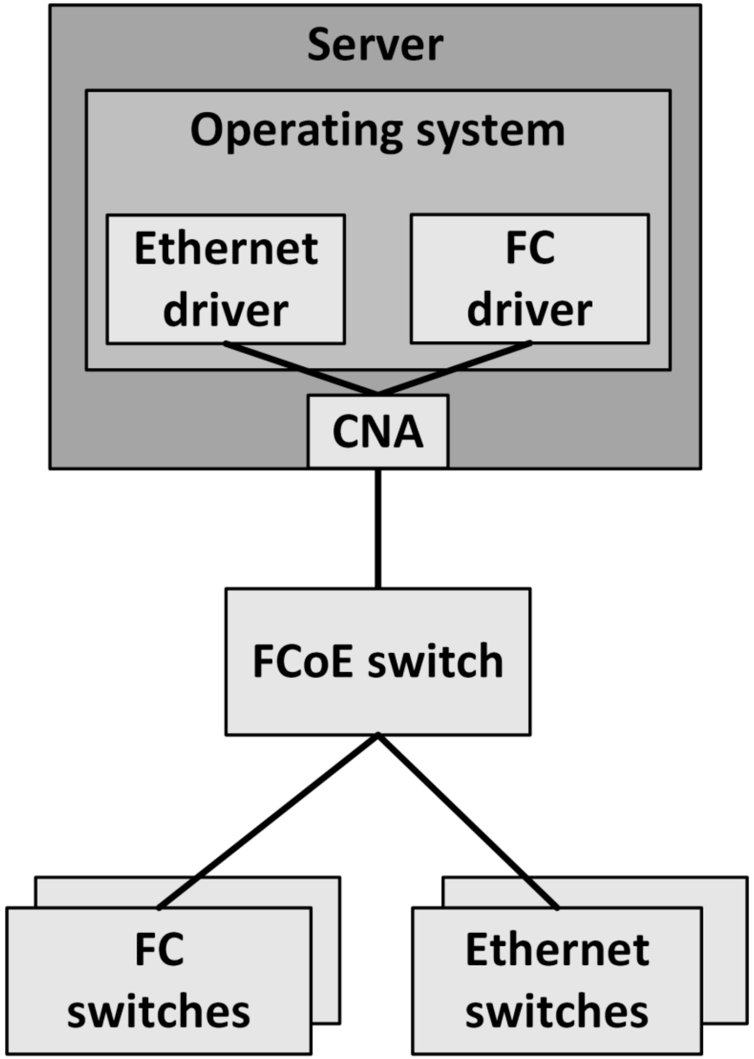
- Fibre Channel over Ethernet (FCoE) encapsulates Fibre Channel data in Ethernet packets
- Allows Fibre Channel traffic to be transported over 10 Gbit or higher Ethernet networks
- FCoE eliminates the need for separate Ethernet and Fibre Channel cabling and switching technology
- PCoE needs at least 10 Gbit Ethernet with special extensions, known as Data Center Bridging (DCB) or Converged Enhanced Ethernet (CEE)

FCoE

- Ethernet extensions:
  - Lossless Ethernet connections
    - A FCoE implementation must guarantee that no Ethernet packets are lost
  - Quality of Service (QoS)
    - Allows FCoE packets to have priority over other Ethernet packets to avoid storage performance issues
  - Large Maximum Transfer Unit (MTU) support
    - Allows Ethernet packets of 2500 bytes in size, instead of the standard 1500 bytes
    - Also known as Jumbo frames

FCoE

- FCoE needs specialized Converged Network Adapters (CNAs)
- CNAs support the Ethernet extensions
- They present themselves to the operating system as two adapters:
  - Ethernet Network Interface Controller (NIC)
  - Fibre Channel Host Bus Adapter (HBA)



iSCSI

- iSCSI allows the SCSI protocol to run over Ethernet LANs using TCP/IP
- Uses the familiar TCP/IP protocols and well known SCSI commands
- Performance is typically lower than that of Fibre Channel, due to the TCP/IP overhead
- With 10 or 40 Gbit/s Ethernet and jumbo frames, iSCSI is now rapidly conquering a big part of the SAN market

Network Attached Storage (NAS)

- Also known as a File Server, is a network device that provides a shared file system to operating systems over a standard TCP/IP network
- It implements the file services and holds the disks on which data is stored
- Can use external disk storage provided by a SAN
- Can provide snapshot and clone technology at a file level

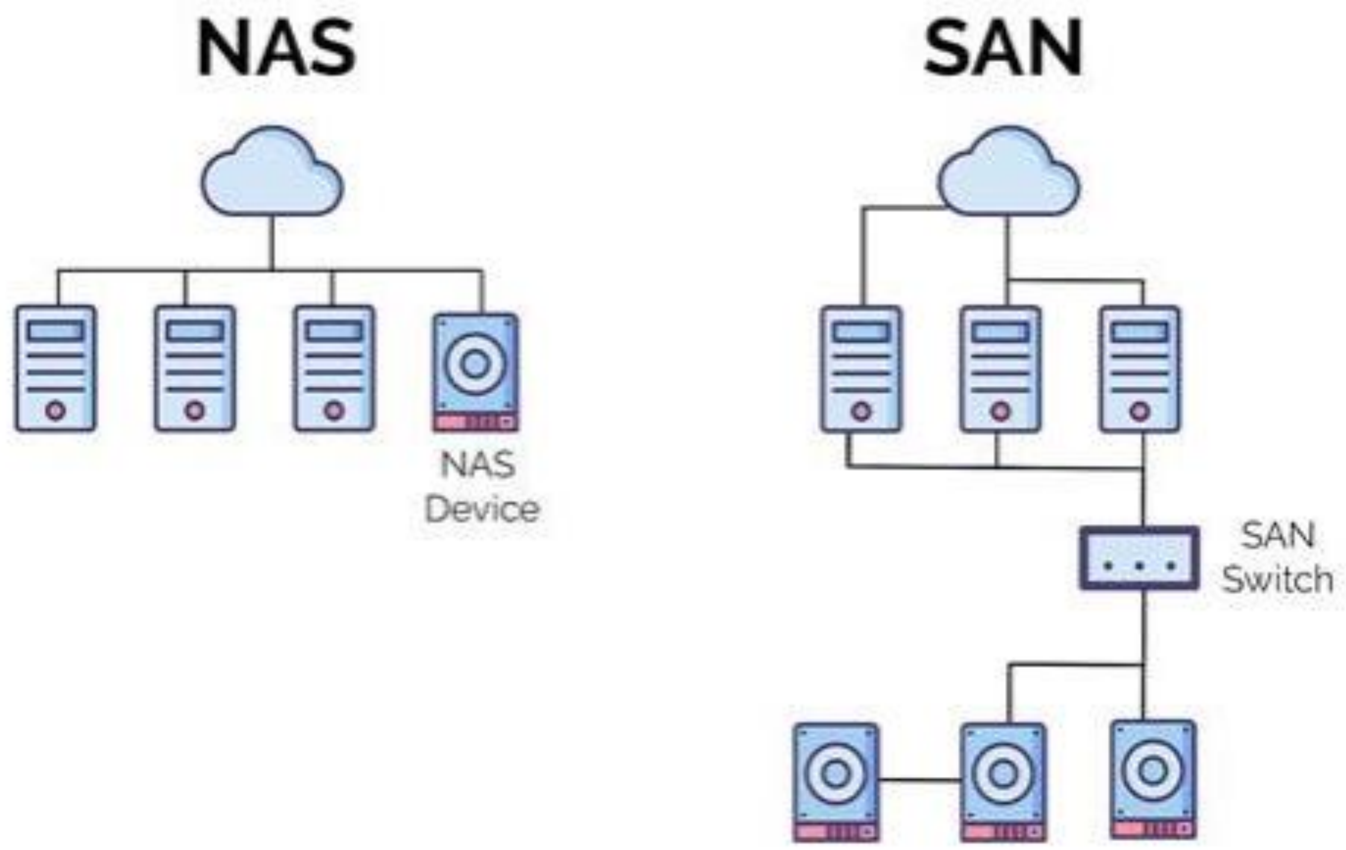
Network Attached Storage (NAS)

- The difference between a SAN and NAS:
  - SAN:
    - Offers **disk blocks** (unformatted disks called LUNs) that can be **used by only one server**
    - Uses iSCSI, Fibre Channel or FCoE as the communication layer
  - NAS:
    - Offers a **shared filesystem** to store files that can be **used by multiple servers**
    - Connects to for instance to an LDAP or Active Directory service in order to set file and/or folder permissions
    - Uses SMB/CIFS or NFS over TCP/IP as the communication layer

Network Attached Storage (NAS)

- A clustered NAS is a NAS that uses a distributed file system running simultaneously on multiple servers
  - Distributes data and metadata across storage devices
  - Still provides unified access to the files from any of the cluster nodes, unrelated to the actual location of the data

- File shares can also be provided by public cloud providers
  - AWS offers File Gateways
  - Azure has Storage Accounts
  - GCP has Filestore


NAS
NAS
Device
SAN
SAN
Switch

Object Storage

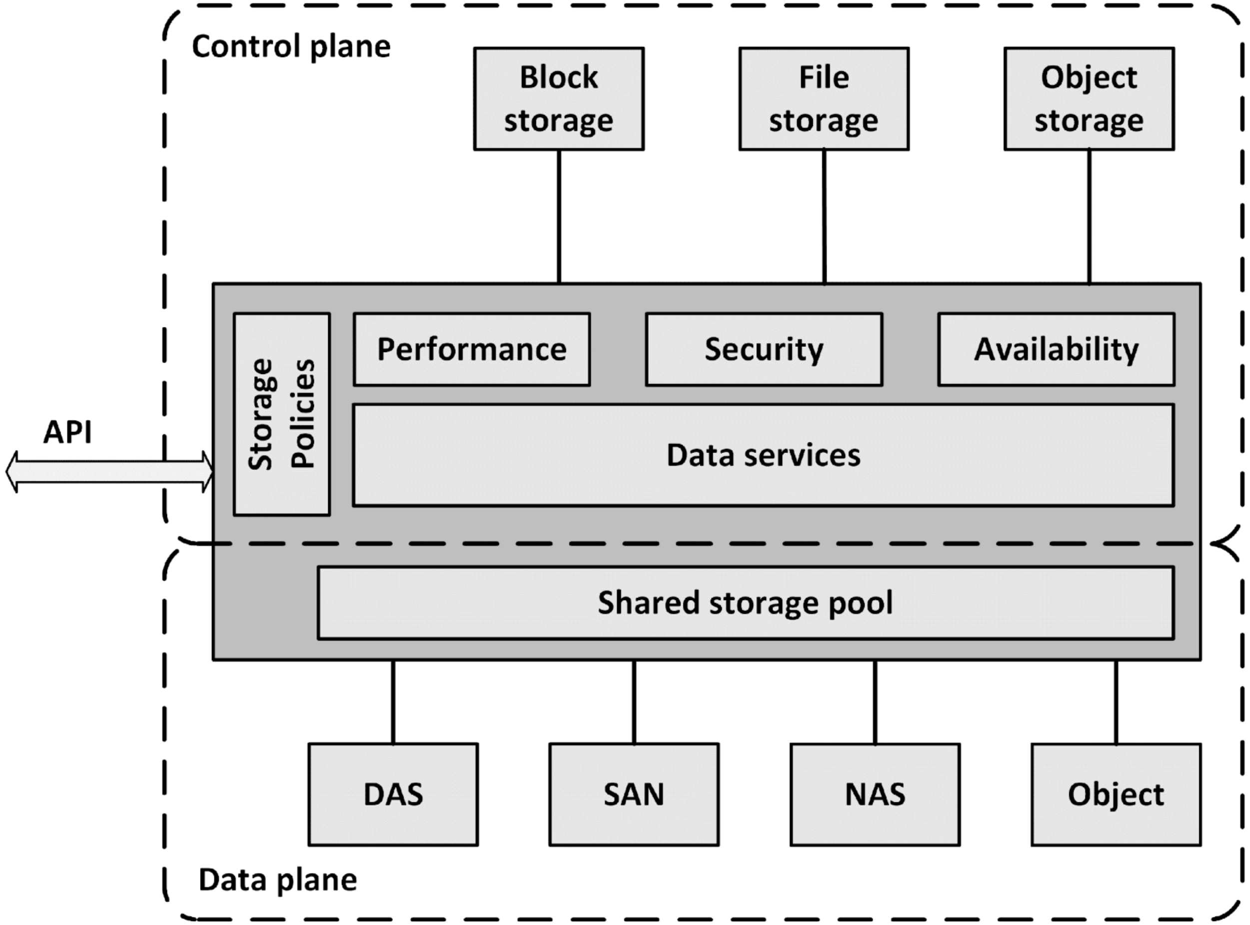
- It manages **data as objects**
- It stores and retrieves data using a **REST API** calls over HTTP, is served by a webserver, and is designed to be highly scalable
- All large public cloud providers offer object storage services
  - AWS has S3
  - Azure has Blob Storage as part of storage accounts
  - GCP provides Object Storage
- These services are **massively scalable** and data **is stored in multiple locations** geographically to ensure no data will be lost in case of a disaster

Object Storage

- Data in object storage **can’t be modified**. If a file is modified, the original file is deleted, and a new file is created
- It is a **good fit for data that doesn't change much**, like office documents, backups, archives, video and audio files, and virtual machine images
- Some systems emulate a file system using object storage
  - For instance, Amazon’s S3FS creates a virtual filesystem, based on S3 object storage, that can be mounted to an operating system in the traditional way, however, with significant performance degradation
  - A much better solution is to use object storage with applications designed for it

Software Defined Storage

- Software Defined Storage (SDS) abstracts data and storage capabilities (also known as the control plane) from the underlying physical storage systems (the data plane)



Software Defined Storage

- SDS virtualizes all physical storage into one large shared storage pool
- Storage can be implemented as software running on commodity x86-based servers with direct attached disks
- Physical storage can also be a SAN, a NAS, or an Object storage system
- The storage offers from all major cloud providers is based on SDS

Software Defined Storage

- From the shared storage pool, software provides data services like:
  - Deduplication
  - Compression
  - Caching
  - Snapshotting
  - Cloning
  - Replication
  - Tiering

Software Defined Storage

- SDS provides servers with virtualized data storage pools

- Example:
  - A newly deployed database server can invoke an SDS policy that mounts storage configured to have its data striped across a number of disks, creates a daily snapshot, and has data stored on tier 1 disks

- APIs can be used to provision storage pools and set the availability, security and performance levels of the virtualized storage

- Using APIs, storage consumers can monitor and manage their own storage consumption

Storage availability

Storage availability

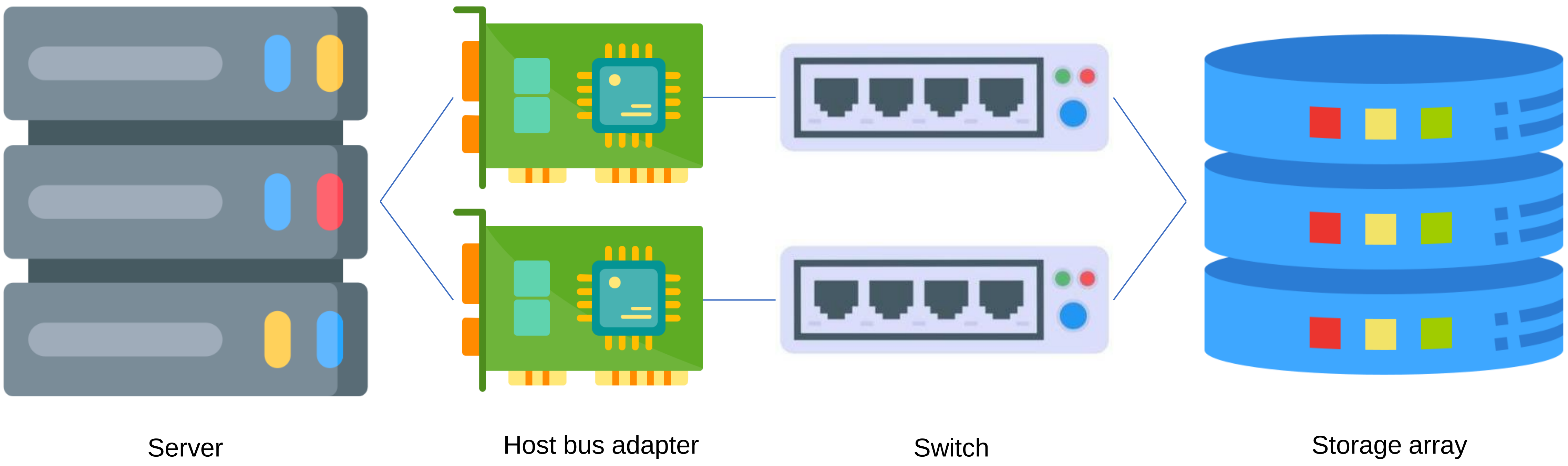
- Storage availability refers to the **uptime** of your storage system.

- How often it's accessible for storing, retrieving, and managing your data?

Redundancy and data replication

- To increase availability in a Storage Area Network (SAN), components like **host bus adapter** (HBA)s and **switches** can be installed redundantly
- Using **multiple paths** between **HBAs** and SAN **switches**, failover can be instantiated automatically when a failure occurs
- Multiple storage systems can be used. Using replication, changed disk blocks from the primary storage system are continuously sent to the secondary storage system, where they are stored as well

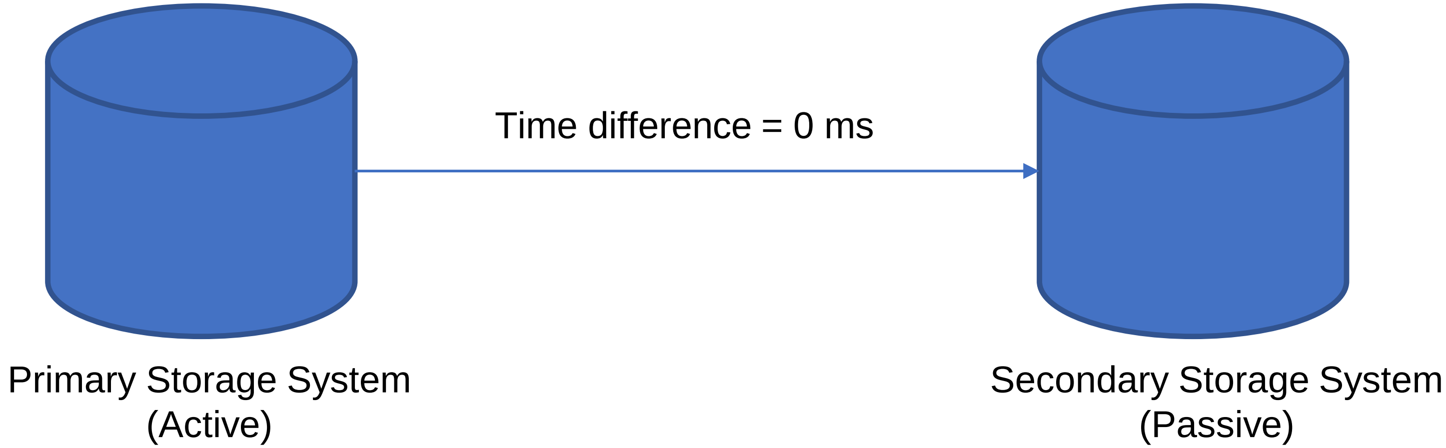
Host Bus Adapter (HBA)


Server
Host bus adapter
Switch
Storage array

Redundancy and data replication

<u>Synchronous replication</u>

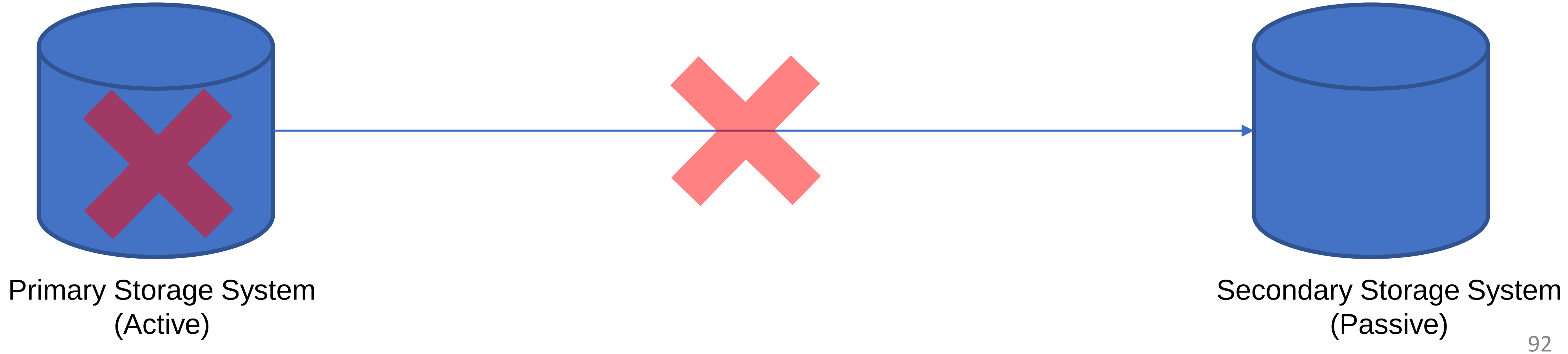
- Each write to the **active storage system** and the replication to **the passive storage system** must be completed before the write is confirmed to the operating system
- Ensures data on both storage systems is synchronized at all times and **data is never lost**



Redundancy and data replication

Synchronous replication

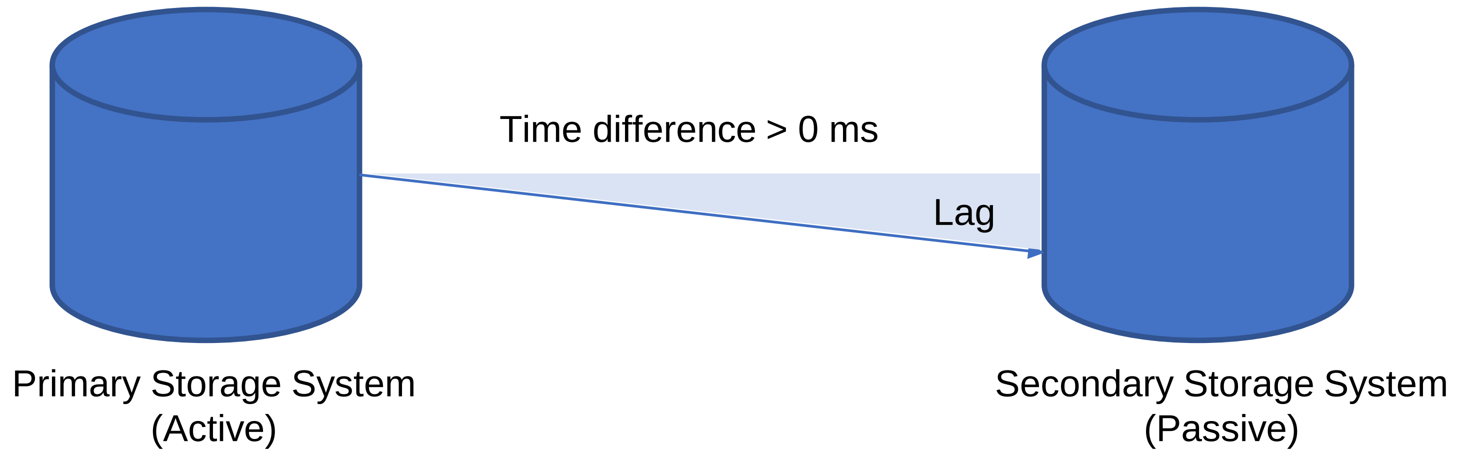
- When the physical cable length between the two storage systems is more than 100 km, latency times get too long, slowing down applications, that have to wait for the write on the secondary storage system to finish
- Risk: a failing connection between both storage systems a write is never finished, as the data cannot be replicated. This effectively leads to downtime of the primary storage system



Redundancy and data replication

Asynchronous replication

- After data has been written to the primary storage system, the write is immediately committed to the operating system, without having to wait for the secondary storage array to finish its writes as well



Redundancy and data replication

Asynchronous replication

- Asynchronous replication does **not have the latency impact** that synchronous replication has

- Disadvantage: **potential data loss** when the primary storage system fails before the data has been written to the secondary storage system

Backup and recovery

- **Replication does not protect** against **data erasure** or **data integrity problems** due to a software bug. If this happens, the deleted or altered data will also be replicated to the secondary storage system

- Backups are copies of data, used to restore data to a previous state in case of data loss, data corruption or a disaster recovery situation

- Backups are always a last resort, only used if everything else fails, to save your organization in case of a disaster

- A well-designed system should have options to repair incorrect data from within the system or by using systems management tools (like database tools)

Backup and recovery

- In general, backups should not be kept for a long time
  - Because the data copies are only relevant in the event of a disaster, organizations will typically have little use to restore a data backup that is more than a few weeks old
  - Restoring a backup takes you back in time. Like a time machine, but without the rest of the world – like your business partners and customers – going back in time as well

Backup and recovery

- A common mistake is to mix up backup with archiving
  - **Backup** is **about protection against data loss**
  - **Archiving** deals with **long term data storage**, in order to comply with law and regulations

- Backups should not be used to view the status of information from the past
  - It should be possible to retrieve these statuses from the system itself
  - **No data should ever be deleted** in a typical production system
  - Older data could be archived to a secondary system or database

Backup and recovery

- Backups need to be made at a regular basis
  - Usually daily
  - Sometimes more often – every hour, or even continuously in highly critical environments

- 3-2-1 rule:
  - Keep **three copies** of your data
  - on **two** different **media** types
  - with **one** copy stored at a **separate location**

Backup and recovery

- Backups must be available at a secondary site for restore
  - Experience with real world disasters shows it is good practice to have a distance of at least 5 km between the main site and the backup data

- Apart from application data, a copy must be available on the secondary site of:
  - Operating system installation disks
  - Printed procedures on how to build up a new system using the backups
  - License keys of the software (including the restore software)

Backup and recovery

- Test the restore procedure at least once a year to ensure restores work as planned
  - Include building up new hardware
  - Have restore procedures tested by a third party, or at least by people that have not performed a restore before
  - In case of a real disaster we cannot assume that systems managers are able to restore data again
- Restore tests should be performed each month to ensure backup media still work as expected
  - Restore some files
  - Do the tapes really contain the expected data?

Backup schemes

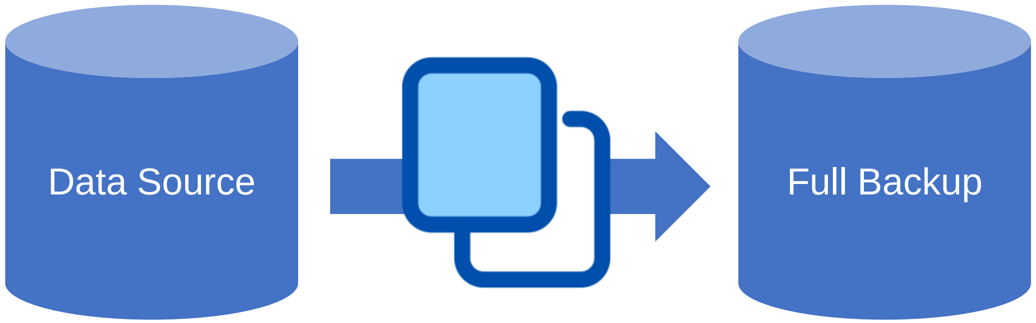
- A backup scheme describes what data is backed-up, when, and how

- Four basic backup schemes:
  1. Full backup
  2. Incremental backup
  3. Differential backup
  4. Incremental forever backup
  5. Continuous data protection (CDP)

Backup schemes

1. Full backup

- A complete copy of all data
- Full backups are only created at relatively large intervals (like a week or a month)
- Creating them takes much time, disk or tape space, and bandwidth
- Restoring a full backup takes the least amount of time



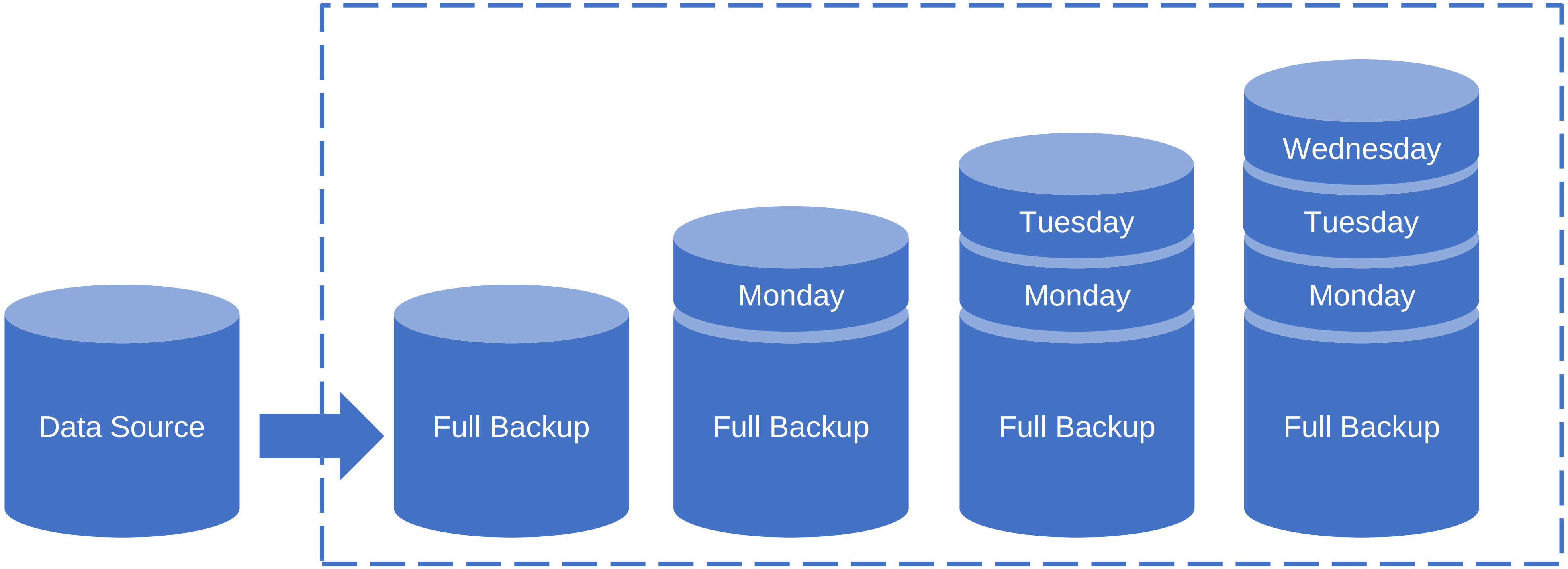
Backup schemes

2. Incremental backup

- **Small backup file :** save only newly created or changed data since the last backup, regardless of whether it is a **previous incremental backup** or **a full backup**
- **Restoring** an incremental backup can **take a long time**, especially when the last full backup is many incremental backups ago

Backup schemes

2. Incremental backup



Backup schemes

3. Differential backup

- Save only **newly created or changed data** since the **last full backup**

- Restoring a differential backup is quite efficient, as it implies restoring a full backup and only the most recent differential backup

Backup schemes

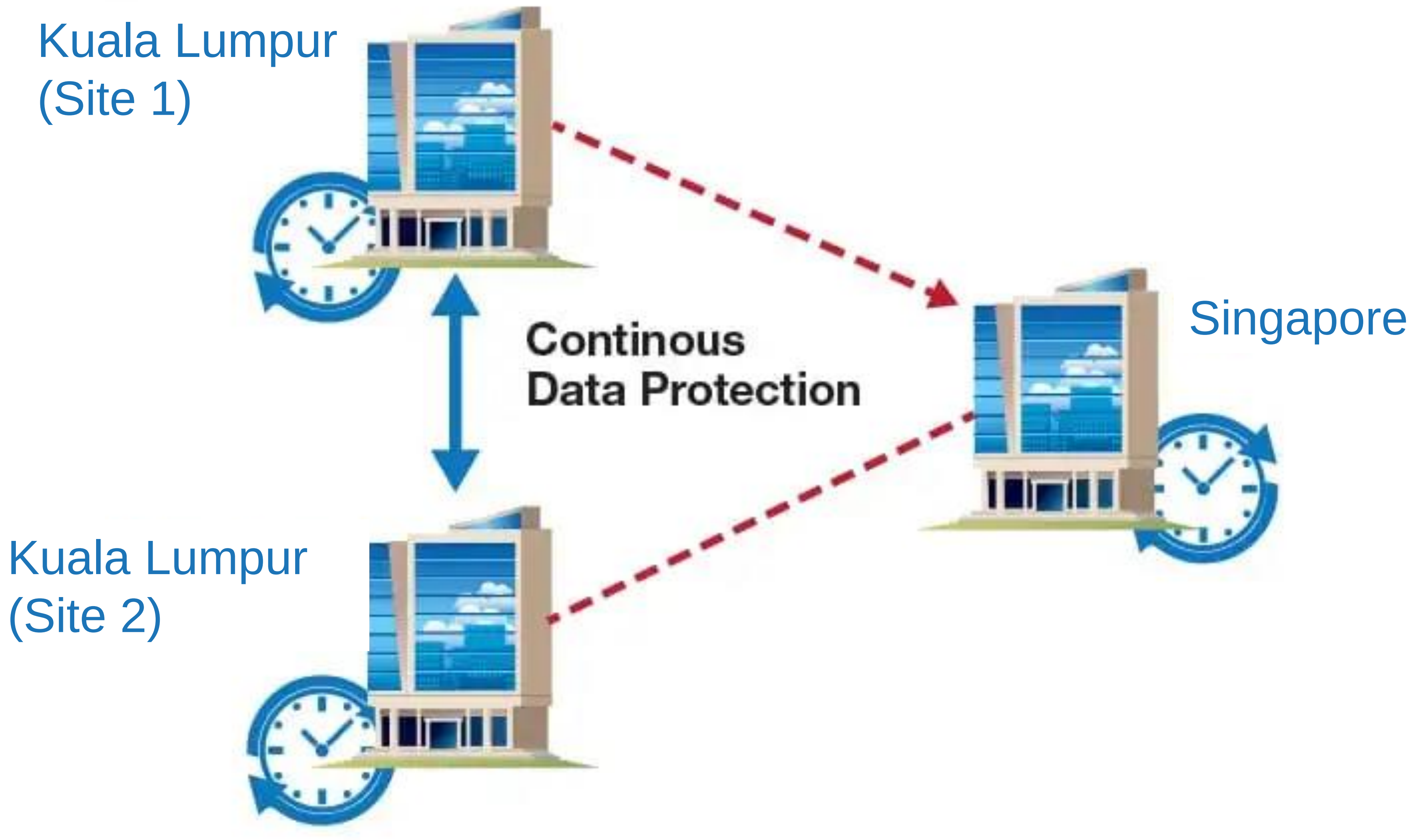
4. Incremental forever backups

- Make an **initial full backup**, after which only **incremental backups** are sent to the backup system
- Metadata about the increments are stored in the backup system, allowing the backup system to compile a point in time restore from the increments

Backup schemes

5. Continuous Data Protection (CDP)

- Guarantees that every change in the data is also simultaneously made in the backup system
- The RPO (Recovery Point Objective) is set to zero, because each change immediately triggers a backup process
- Expensive technology, and therefore only used in specific situations


Kuala Lumpur
(Site 1)
Continous
Data Protection
Singapore
Kuala Lumpur
(Site 2)

Backup data retention time

- Backup data retention time is the amount of time in which a given set of data will remain available for restore

- Defines how long backups are kept and at which interval

Backup data retention time

- In practice, a Grandfather-Father-Son (GFS) based schedule is often used:
  - Each day a backup is made
  - After a week, there are seven backups, of which the oldest backup is renamed to a weekly backup
  - After the second week, the same is done and the daily backups of the week before are deleted
  - Now there are eight backups: seven daily, two weekly
  - Every four weeks, the weekly backup is renamed as a monthly backup and the weekly backups are reused
  - The daily backups are the son, the weekly backups are the father, and the monthly backups are the grandfather

Grandfather

Monthly

Father

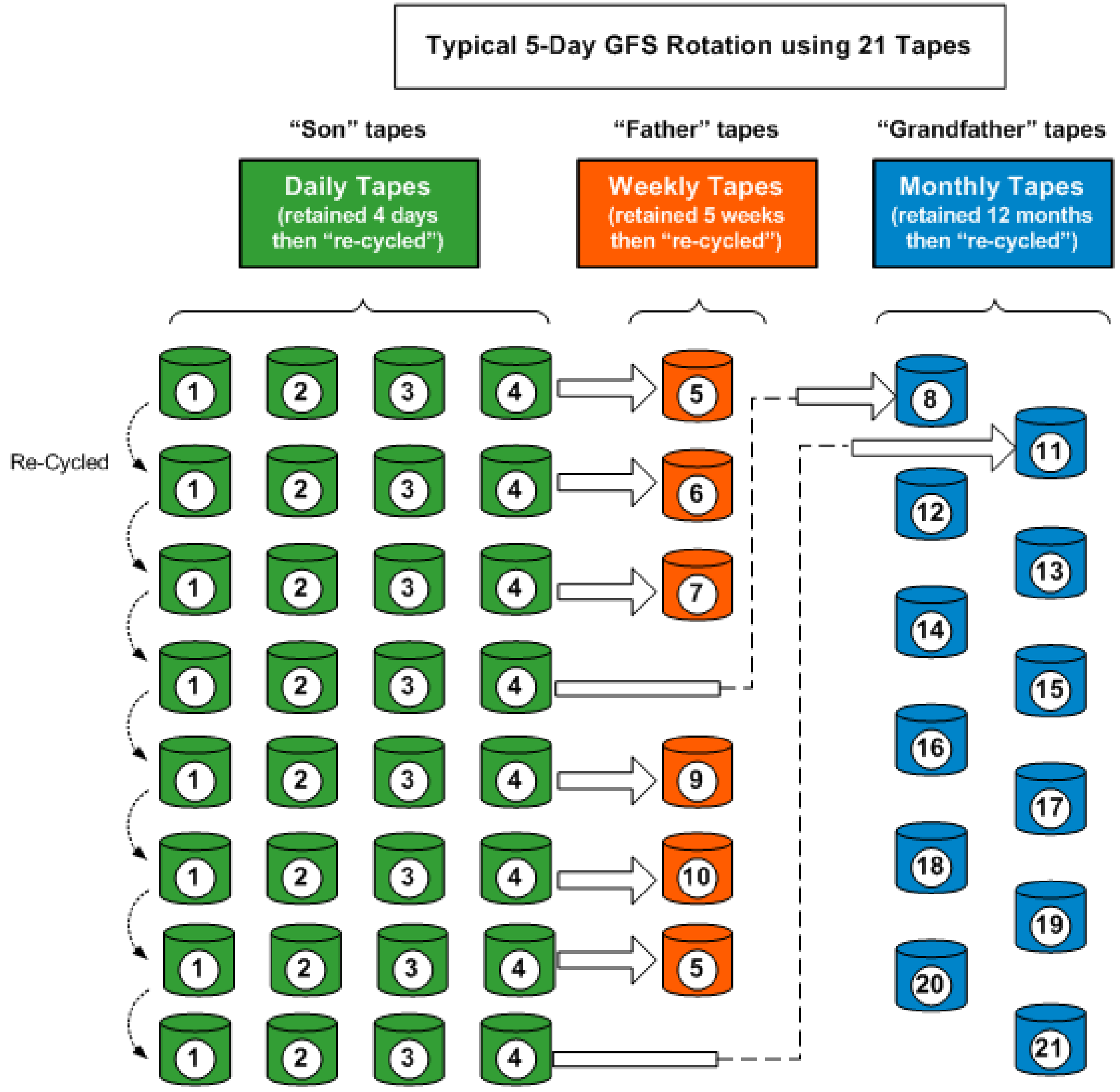
Weekly

Son

Daily

Now

Now


Typical 5-Day GFS Rotation using 21 Tapes
"Son" tapes
"Father" tapes
"Grandfather" tapes
Daily Tapes
(retained 4 days then "re-cycled")
Weekly Tapes
(retained 5 weeks then "re-cycled")
Monthly Tapes
(retained 12 months then "re-cycled")
Re-Cycled

Archiving

- Archiving is mostly done for **compliancy** and **regulation** reasons

- Example:
  - US regulations require all medical records to be retained for 30 years after a person's death
  - This means that X-rays taken when a child was born must be kept for as much as 130 years!

- Noncompliance to law and regulation can lead to serious business disruption, fines, and even jail time

HASIL

11.1 Retention Of Records

11.1.1 Taxpayers are required to keep sufficient records for a period of seven years from the end of the year to which income from the business relates, as provided under paragraph 82(1)(a) of the Act, to enable the DGIR to ascertain income or loss from the business. Subsection 82(8) further provides that all records relating to any business in Malaysia must be kept and retained in Malaysia. 'Records' under subsection 82(9) include books of accounts, invoices, vouchers, receipts and other documents necessary to verify entries in any books of accounts.

Source: https://www.hasil.gov.my/en/international/transfer-pricing/chapter-xi-documentation/111-retention-of-records/

How long do I need to keep the records?

The records must be kept for at least 6 years following the date of completion of the transaction or termination of the business relationship or occasional transaction.

The records must be kept in a form that is admissible as evidence in court pursuant to the Evidence Act 1950.

Source: https://amlcft.bnm.gov.my/record-keeping

Archiving

- Archived data is **read-only** to protect it from being altered
  - Very important for regulatory compliance and non-repudiation
  - Some archiving systems store data in an encrypted form and use digital signatures to prove data is not tampered with
  - Some systems allow data to be written to it for archiving, but disallow changing or deleting data
    - CD / DVD/ Blu-ray
    - WORM tapes

Archiving

- Data must be kept in such a way that it is guaranteed the data can be read after a long time
  - Digital format (like a Microsoft Word file or a JPG file)
  - Physical format (like a DVD or a magnetic tape)
  - Storage environment (temperature, humidity)

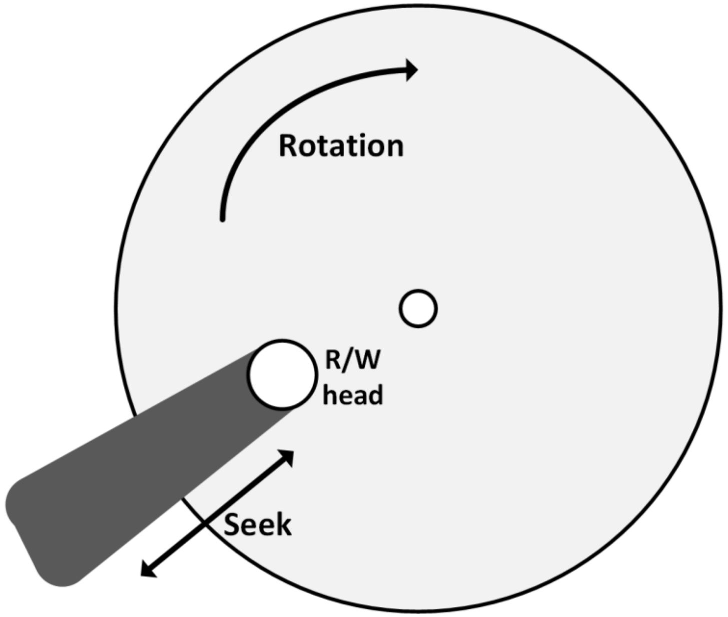
Archiving

- Use open standards for storing archived data
  - Open standards are well documented
  - Reading data will always be feasible, using emulation software if needed
  - Storing all documents in structured human-readable XML text files is one way to ensure data can be read for many decades
- Transfer data that is to be kept for a long time to the latest storage media standard every 10 years

Storage performance

Disk performance

- Disk performance is dependent on:
  - Disk rotation speed
  - Seek times
  - Interface protocol



- Some common examples of rotation delay:

Disk RPM	Average rotational delay (ms)
5,400	5.6
7,200	4.2
10,000	3
15,000	2

Disk performance

- Disks cannot spin much faster than 15,000 RPM
  - At this speed the velocity at the edge of a 3.5” disk is 250 km/h!
  - Increasing this velocity would physically destroy the disk
- Seek time is the time it takes for the head to get to the right track
  - Average seek times:
    - 3 ms for high-end disks
    - 9 ms for low-end disks

IOPS

- Input/output Operations Per Second (IOPS) is a measure of how many read and write operations a disk can complete in one second
- $\frac{1000}{\text{Rotational delay (ms)} + \text{Seek time(ms)}}$

- Writing is typically a bit slower than reading

- Typical IOPS:

Disk type	IOPS
Mechanical disks	< 500
SSD	400,000 read 150,000 write
NVMe	1,000,000 read 130,000 write

RAID penalty

- In RAID sets multiple disks are used to form one virtual disk (LUN)
- Writing data on multiple disks introduces some delay, known as the RAID penalty
- Penalties for various RAID configurations are
  - RAID 0: no penalty
  - RAID 1: penalty of 2
  - RAID 10: penalty of 2
  - RAID 5: penalty of 4
  - RAID 6: penalty of 6

Interface throughput

- Storage performance is also dependent on how fast the interface can move data from the disks to the systems consuming the data and vice versa

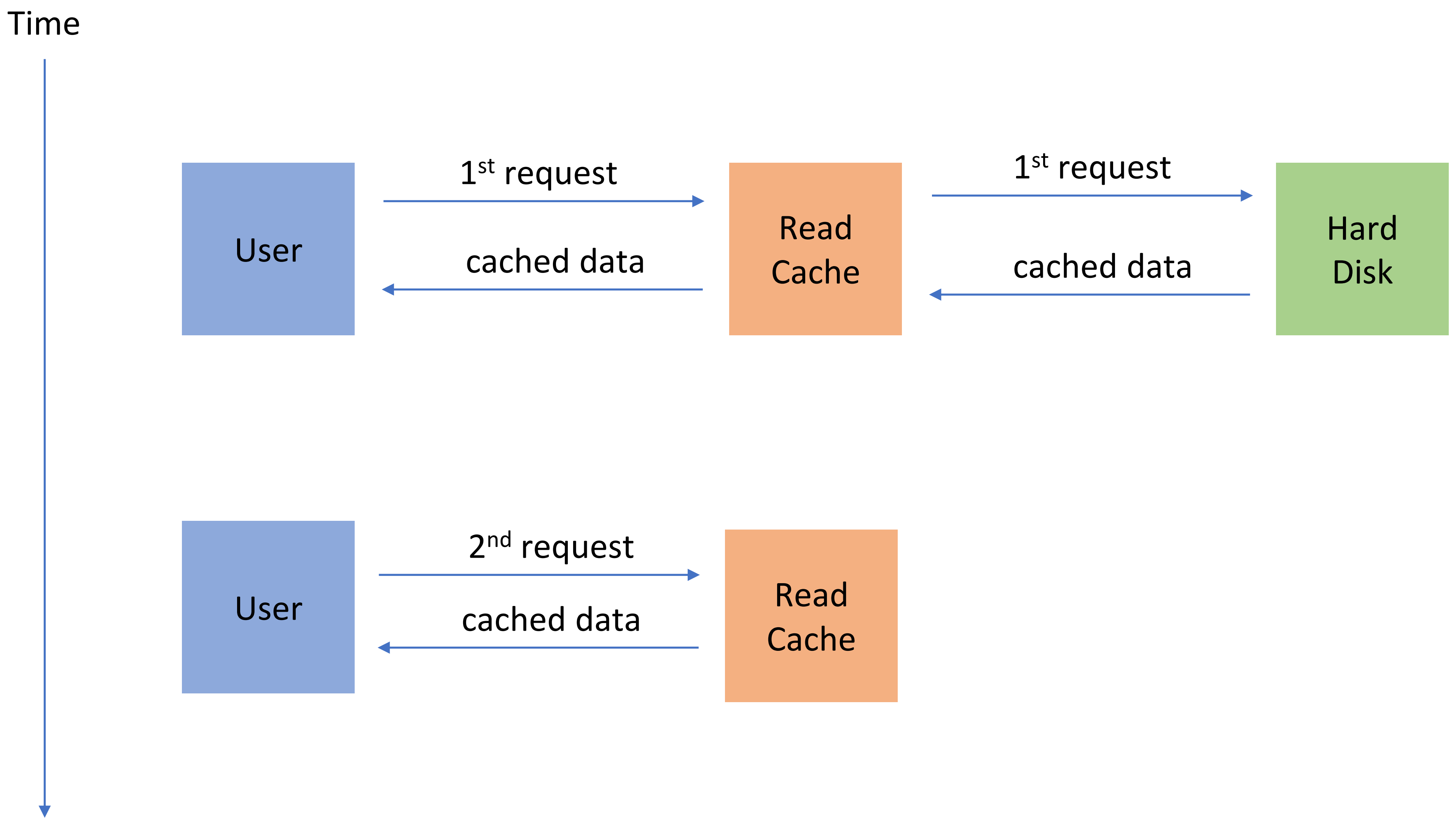
- An overview of the various interface speeds:

Interface	Speed (Gbit/s)	Speed (GB/s)
IDE (Parallel ATA)		0.133
SCSI		0.32
SATA	6	0.768
SAS	12	1.5
NVMe		7
FC	128	16

Caching

- A caching system in disk controllers can improve performance by several orders of magnitude

- **Read-cache** acts as a buffer for reads. When the same data is read multiple times, it is served from cache

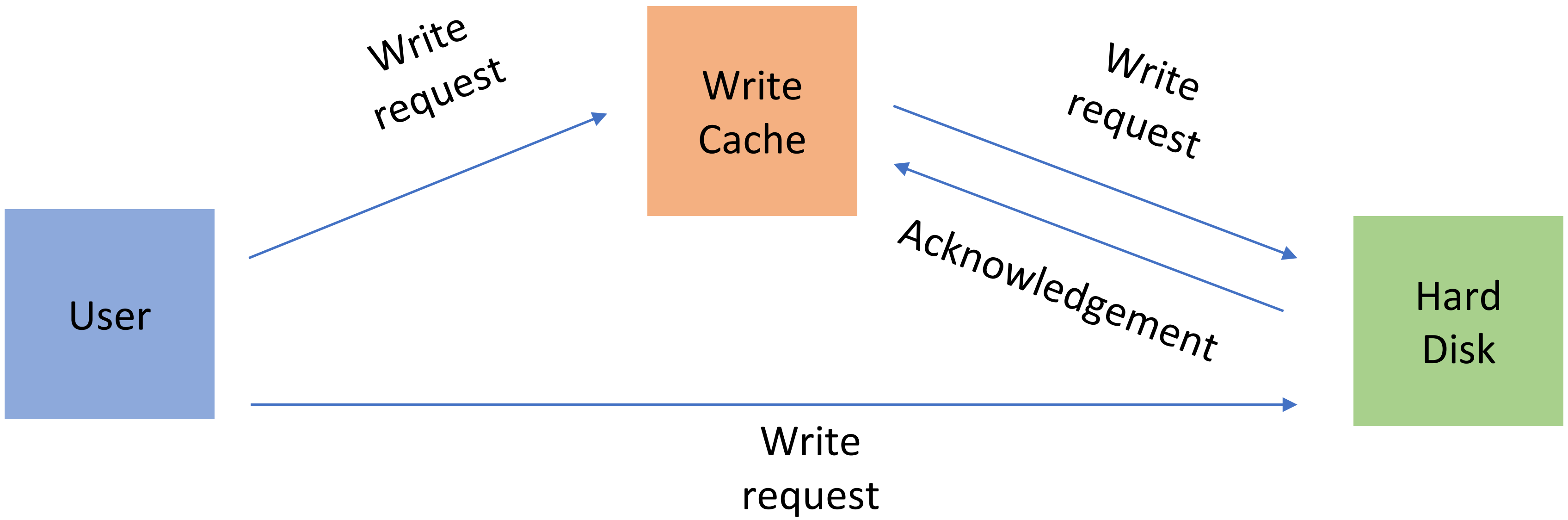

Time
User
1st request
cached data
Read Cache
1st request
cached data
Hard Disk
User
2nd request
cached data
Read Cache

Caching

- **Write-through cache**: data is written to cache and then to disk, and only acknowledged as written when the data is physically written on the disk

- Write-through cache allows the disk controller to acknowledge the data as written as soon as it is held in cache.

- This allows the cache to buffer writes quickly and then write the data to the slower disk when the disk is ready to accept new I/O operations


Write
request
Write
Cache
Write
request
Acknowledgement
User
Hard
Disk
Write
request

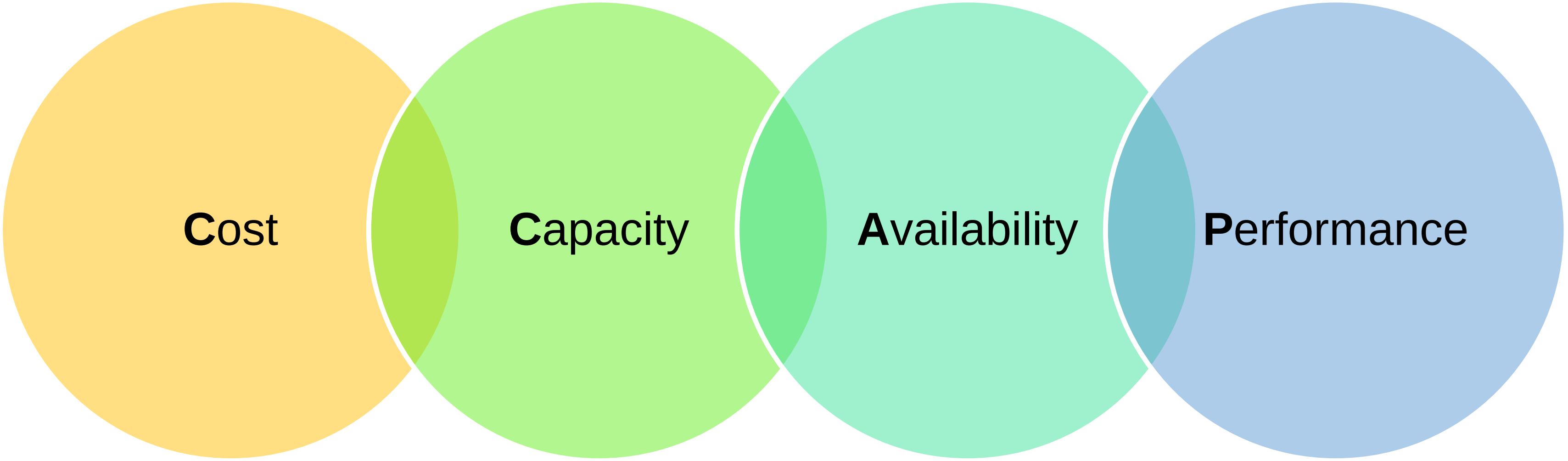
Caching

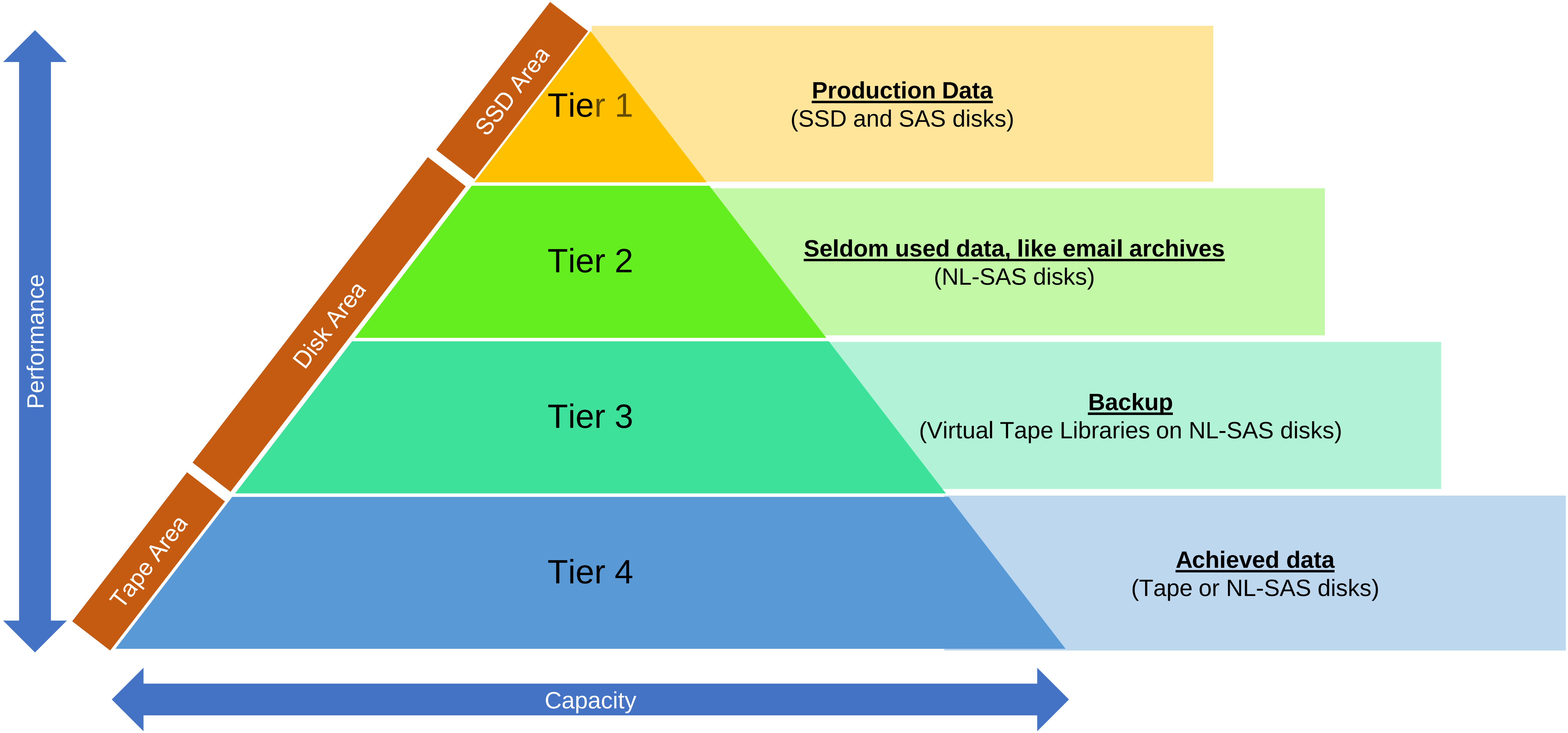
- The type and amount of cache needed depends on what applications need

- A **web server**, for instance, will mostly benefit from **read-cache,** whereas most **databases** are better off with **write cache**

Storage tiering

- Tiered storage creates a hierarchy of storage media, based on cost, performance requirements, and availability requirements





Performance
SSD Area
Disk Area
Tape Area
Tier 1
Tier 2
Tier 3
Tier 4
Production Data
(SSD and SAS disks)
Seldom used data, like email archives
(NL-SAS disks)
Backup
(Virtual Tape Libraries on NL-SAS disks)
Achieved data
(Tape or NL-SAS disks)
Capacity

Storage tiering

- The more tiers are used, the more effort it takes to manage the tiers
- **Automated tiering** usually checks for file **access times**, file **creation date**, and file **ownership**, and automatically moves data to the storage medium that fits best
- Storage tiering is especially **important** when storing data in **the public cloud**, as it has **different storage costs** for each tier

Regions | Dual-regions | Multi-regions

North America | South America | Europe | Middle East | Asia | Africa | Australia

Location	Standard storage (per GB per Month)	Nearline storage (per GB per Month)	Coldline storage (per GB per Month)	Archive storage (per GB per Month)
Taiwan (`asia-east1`)	$0.020	$0.010	$0.005	$0.0015
Hong Kong (`asia-east2`)	$0.023	$0.016	$0.007	$0.0025
Tokyo (`asia-northeast1`)	$0.023	$0.016	$0.006	$0.0025
Osaka (`asia-northeast2`)	$0.023	$0.016	$0.006	$0.0025
Seoul (`asia-northeast3`)	$0.023	$0.016	$0.006	$0.0025
Mumbai (`asia-south1`)	$0.023	$0.016	$0.006	$0.0025

Source: https://cloud.google.com/storage/pricing/?utm_source=google

Load optimization

- **Storage performance** is highly dependent on the **type of load**
- Most vendors recommend a specific storage configuration for their systems or applications
- For example, Oracle recommends a combination of RAID 1 and 5 for its database in order to optimize performance

Storage security

Protecting data at rest

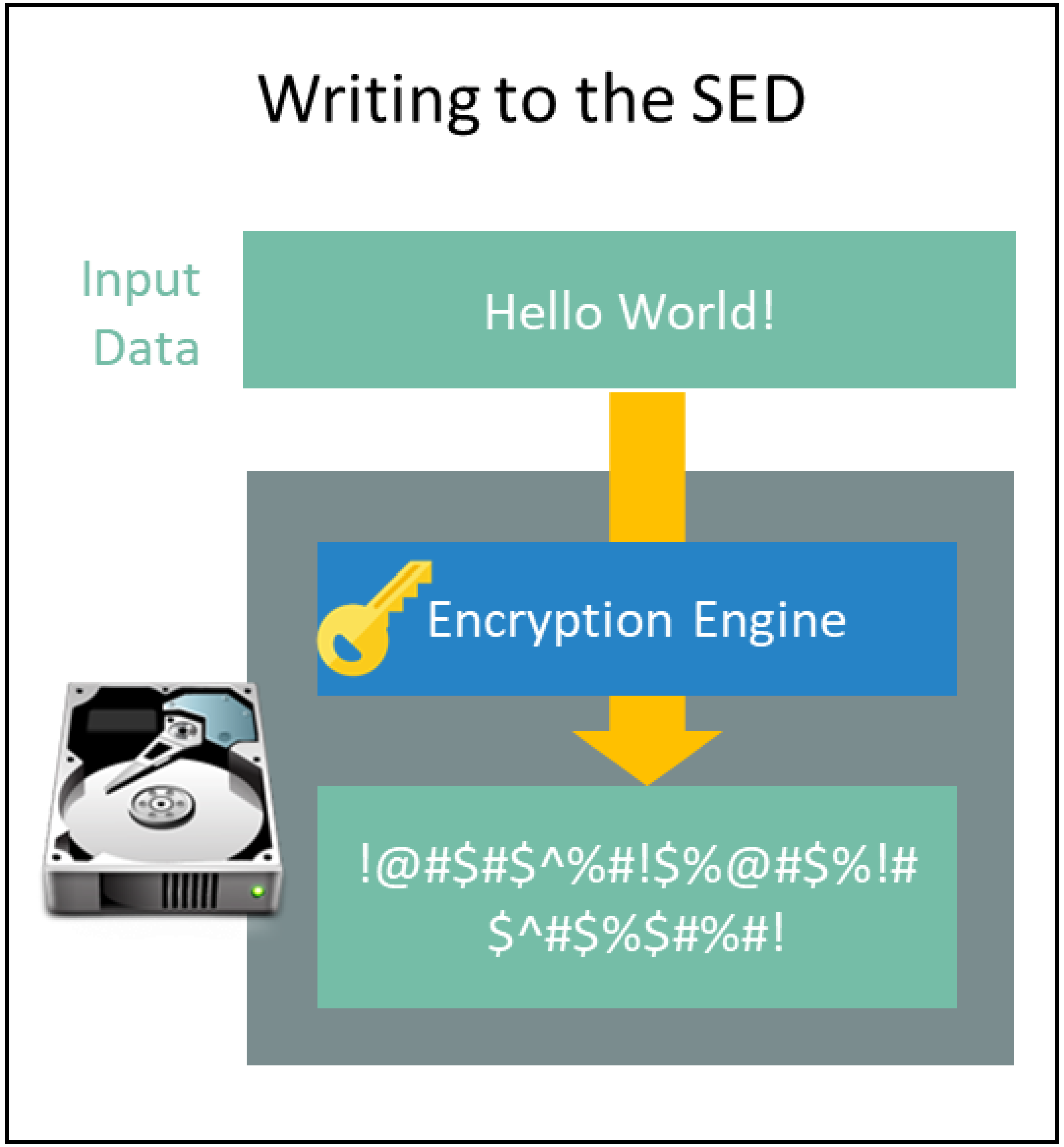
- Data can be in transit, in use, at rest
- Data at rest can be secured using encryption techniques
- Disk encryption in the datacenter has limited benefits:
  - Databases and applications need to work with unencrypted data
  - Disk encryption is only useful when the disks are physically lost or stolen
  - Disks in the datacentre are in a physically secure area

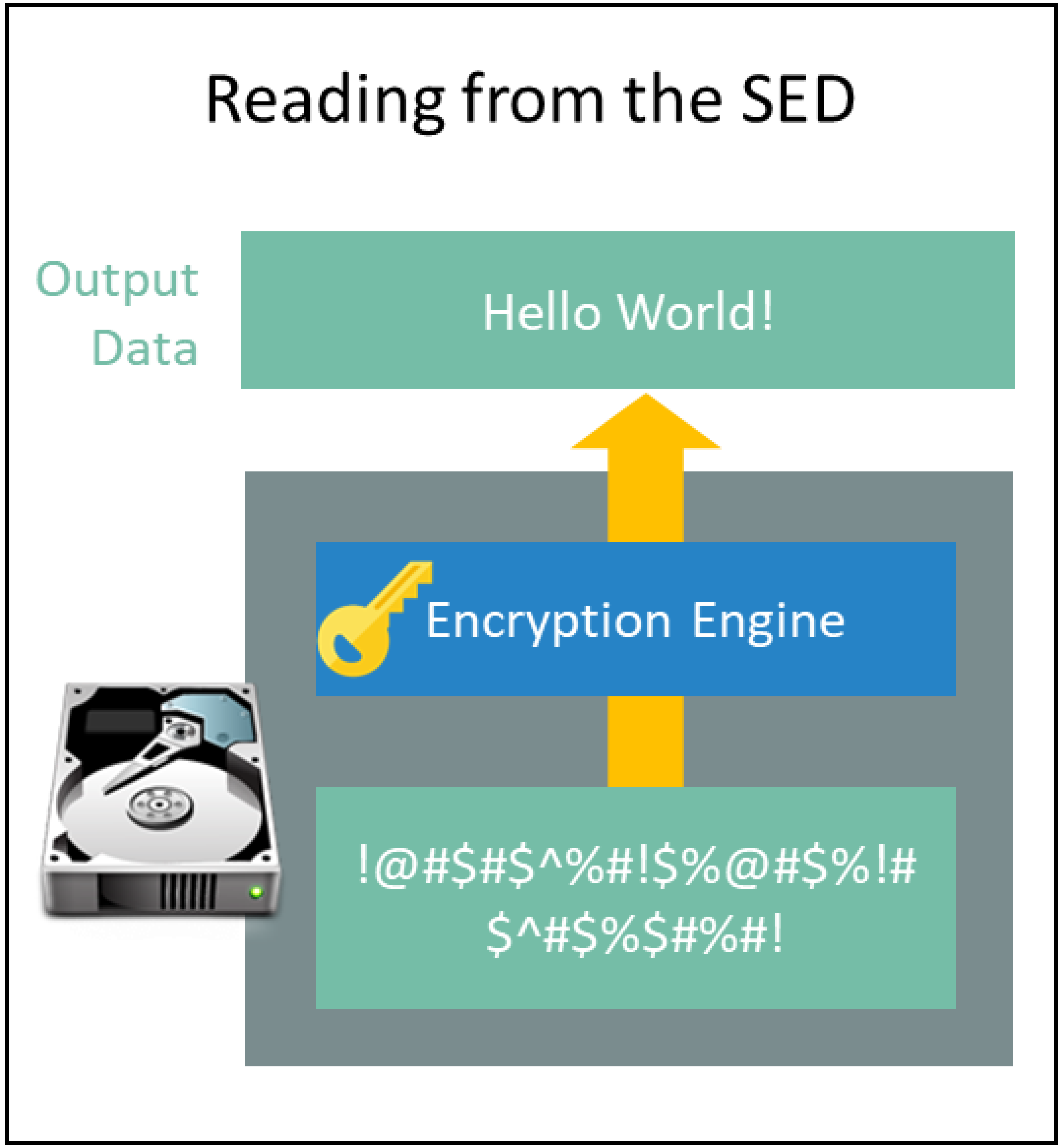
Protecting data at rest

- Disk encryption in the datacenter is useful:
  - A disk drive might get in the wrong hands
  - In case of disk failure, having the data encrypted solves the issue of having potentially sensitive data on a disk that can't be accessed anymore, as it is defective
  - Maintenance contracts often require that a failed disk must be sent back to the vendor after replacing it with a new one. Without disk encryption, returning disks may not be possible since a failed disk cannot be erased anymore.
  - Full disk encryption makes it harder for an attacker to retrieve data from the "empty" space on the disks, which often contains traces of previously stored data.

Protecting data at rest

- Self-Encrypting Drives (SEDs):
  - Use in laptops and desktops
  - When an SED is powered up, authentication is required to access data – the user must type in a password to start the boot sequence of the computer
  - Encryption is built into the disk drive’s hardware
  - Encryption keys are stored on the disk


Writing to the SED
Input Data
Hello World!
Encryption Engine
!@#$#$^%#!$%@#$%!#
$^#$%$#%#!


Reading from the SED
Output Data
Hello World!
Encryption Engine
!@#$#$^%#!$%@#$%!#
$^#$%$#%#!

Protecting data at rest

- Cryptographic Disk Erasure (CDE):
  - Deletes the encryption key on the disk
  - This has the same effect as erasing all disk contents
  - Without the key, unencrypted data can no longer be read from the disk
  - One of the best ways to fully wipe a disk’s contents

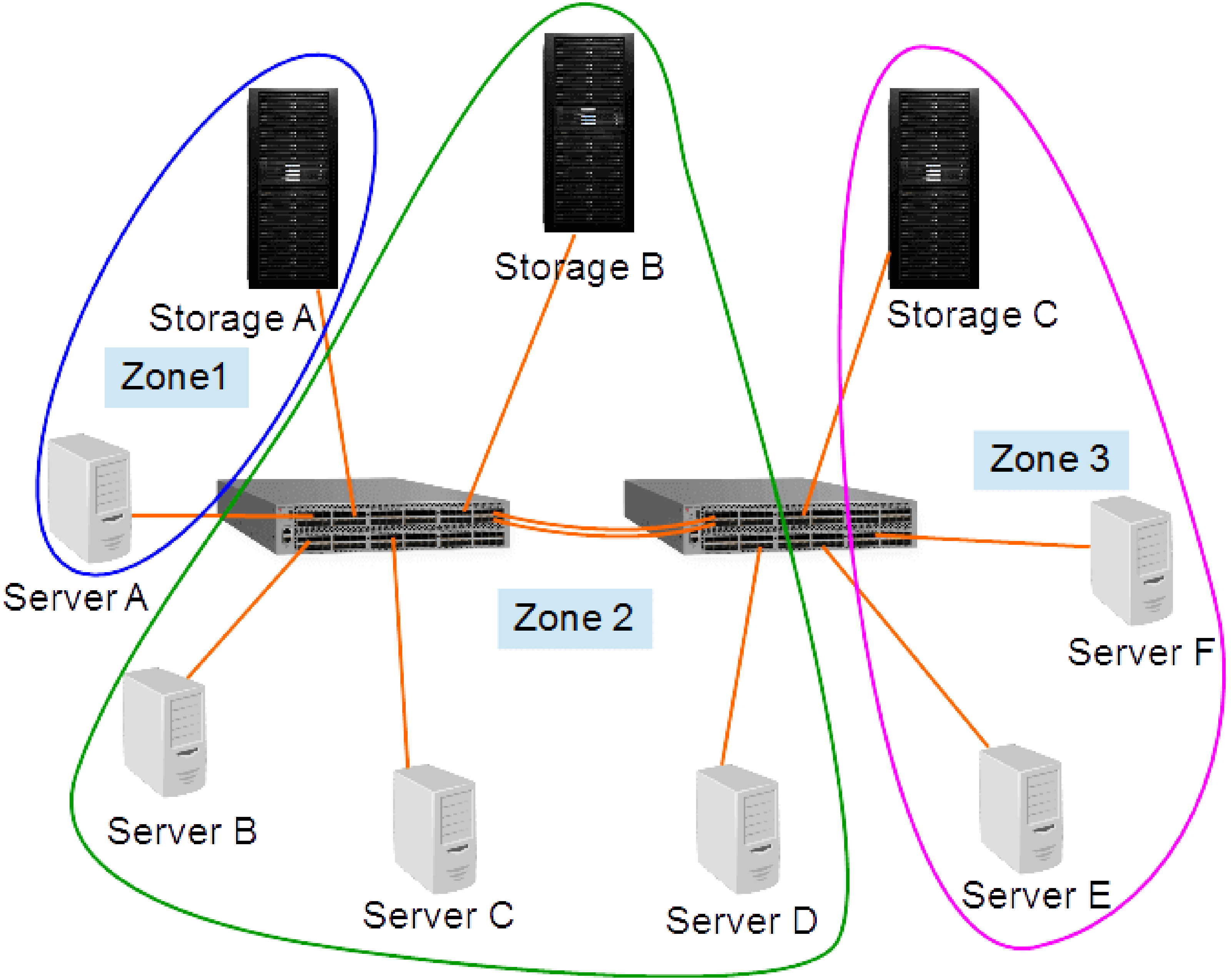
SAN zoning

- SAN zoning is a method of arranging Fibre Channel devices into logical groups on a SAN fabric for security purposes

  - SAN zoning is implemented in the SAN switches

  - SAN zones are comparable with VLANs in Ethernet networks

  - Fibre Channel devices can only communicate with each other if they are members of the same zone



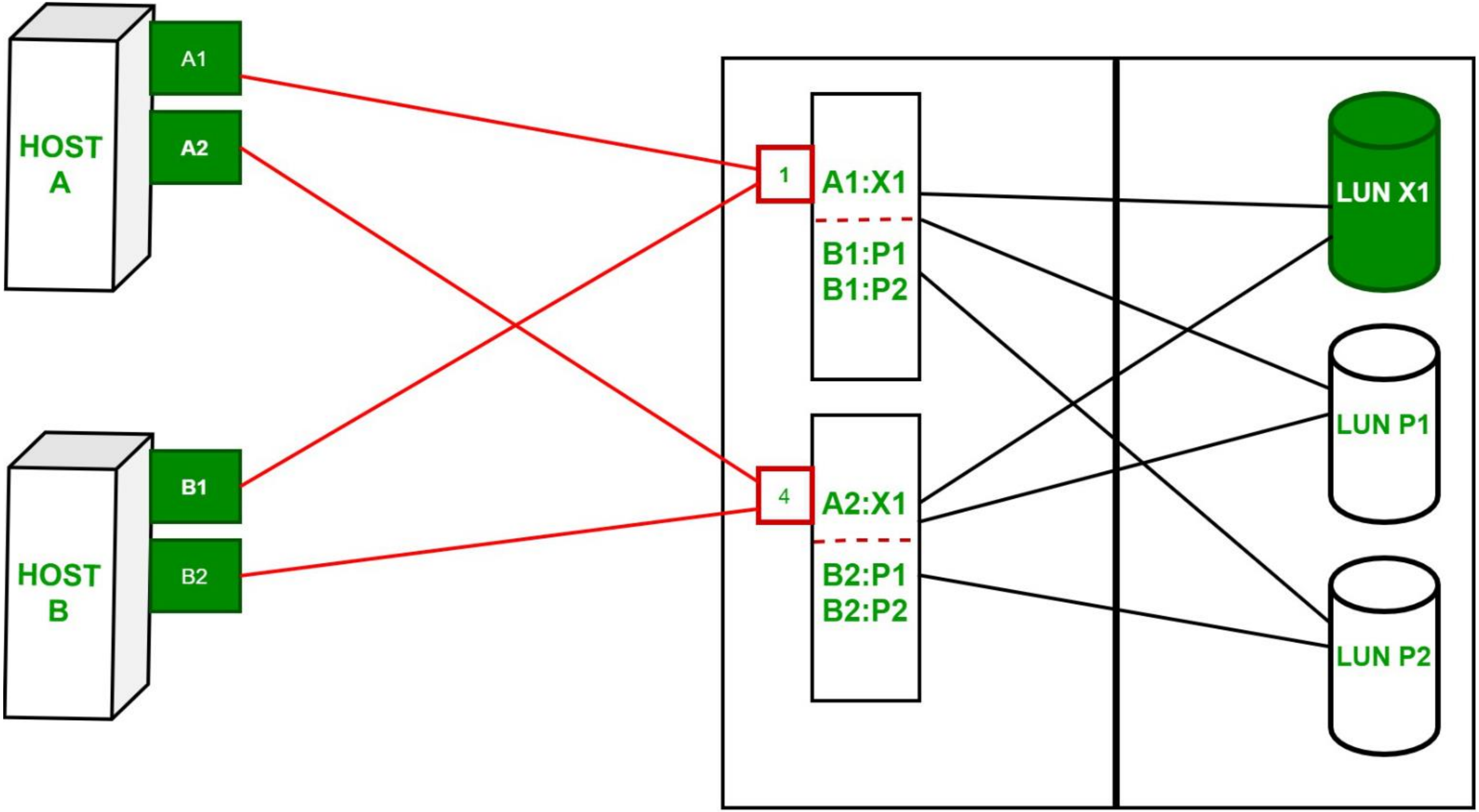
Storage A
Storage B
Storage C
Zone1
Zone 3
Server A
Zone 2
Server F
Server B
Server C
Server D
Server E

SAN LUN masking

- In a SAN, LUN masking makes a LUN available to some hosts and unavailable to other hosts

- LUN masking is implemented primarily at the HBA level, not in the SAN switches

- It is good practice to use a combination of SAN zoning and LUN masking


A1
A2
HOST A
B1
B2
HOST B
1
A1:X1
B1:P1
B1:P2
4
A2:X1
B2:P1
B2:P2
LUN X1
LUN P1
LUN P2

Conclusion

- Discuss the storage building blocks
- Discuss storage availability
- Discuss storage performance
- Discuss storage security